

**PRESERVING PEAT SWAMP FOREST ECOSYSTEM
THROUGH REDD+ ACTIVITY
IN KAMPAR PENINSULA
RIAU-INDONESIA**



Document Prepared By

Korea-Indonesia FMU/REDD+ Project Management Unit¹, SG Consulting², and CER
Indonesia³

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Prepared By	Lee Seong Hwan ¹ , Roh Gyeong Min ² , and Muhammad Ridwan ³
Contact	Room 511, Wing B, 5 th Floor, Blok I, Manggala Wanabakti Building Jakarta ,Indonesia Telephone: +62 21 5773 790 e-mail: leesh00317@gmail.com

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1. PROJECT DETAILS

1.1 Summary Description of the Project

The statistics of the fiscal year 2005 provided that a total of 1.79 GtCO₂e of greenhouse gas was emitted by the Republic of Indonesia, with emissions from Land Use Change and Forestry (LUCF) and peat fire comprising 37% and 25% respectively (SNC, 2010). Consequently, the government of Indonesia suggested a goal of 26% emission reduction based on BAU (Business as Usual) by the year 2020 through unilateral actions. In doing so, prevention of deforestation and peat protection are crucial elements.

Kampar peninsula is located in the Riau province of Sumatra island and is one of the major peat lands in Indonesia with the total area of 671,125 ha (Tropenbos, 2011). Due to the formation of industrial plantation aimed for pulp production starting from 2002, most of the area that was once mostly peat swamp forest has been converted (RPHJP KPHP TBS, 2014). Conversion of peat swamp forest into plantation has estimated to have increased the risks of peat subsidence and fire, which ultimately lead to the continued increase of greenhouse gas emission.

Production Forest Management Unit (KPHP in Indonesian) Tasik Besar Serkap (hereafter referred as "KPHP TBS") is located in Riau Province and manages an area of 513,276 ha within the Kampar Peninsula. The organization is under the government of the Riau province and is authorized of establishment of forest management plans, management/supervision of concession-issued areas, forest protection, and profitable activities in designated areas including timber plantation. Area it manages is classified as national forest with production purpose.

Implementation of REDD+ activities (Reducing Emissions from Deforestation and Forest Degradation, and role of Conservation Forest, Sustainable forest management, and Enhancement of Forest Carbon Stock) play pivotal roles in prevention of deforestation and peat land preservation in Indonesia. As mentioned above, emissions from deforestation and peat decomposition is the major source of national emissions. Therefore, a REDD project is being developed to avoid the conversion of some areas of the Kampar Peninsula.

This project aims to reduce greenhouse gas through preventing 14,722.9 ha managed by KPHP TBS from being converted into a timber plantation. The project area consists of swamp forest on top of a thick layer (5-12m) of peat that is currently maintained under a favourable condition. Nevertheless, the issuance of 17 timber plantation concession over the 355,848 ha out of the 531,276 ha, and an investment analysis result indicate that conversion of the remaining national forest into a plantation would be the best investment. These show that the project area is exposed to great risk of conversion into plantation.

Hence, KPHP TBS intends to protect peat swamp forest through this project using REDD+ Avoided Planned Deforestation (APD) mechanism provided by the Verified Carbon Standard (VCS). In particular, the area also stores more than of **24.8** MtCO₂.

e which allows an expected annual reduction of **1.2** million tonnes CO₂e over the next 20 years through prevention of conversion into plantation. In addition, the prevention of peat swamp forest from conversion into plantation would allow this project to contribute towards protection of Indonesian peat swamp forest and achievement of greenhouse gas emission reduction goal for the year 2020.

1.2 Sectoral Scope and Project Type

Sectoral scope of the project is 14 (agriculture, forest and other land-use).

The project is categorized as Avoid Planned Deforestation (APD) on natural forest.

The project is a single project (not a grouped project).

1.3 Project Proponent

Organization name	KPHP Tasik Besar Serkap, Riau Province-Indonesia
Contact person	Apidian Suherdiata
Title	Head Section of Forest Planning, Utilization, and Use
Address	Jalan Pengayoman 1 Pekanbaru Riau Province-Indonesia
Telephone	+62 761 7874 131 / +62 813 6599 3789
Email	Apidian1973@gmail.com

KPHP is a form of Forest Management Unit assigned to perform the following tasks: (i) Forest management and development of forest management plan, (ii) Forest utilization, (iii) Forest land use utilization, (iv) Forest rehabilitation and reclamation, and (v) Forest protection and nature conservation¹.

Aside from the abovementioned tasks, it also has the authority to describe the national, provincial and district forest policies to be implemented, implementing forest management activities, monitor and evaluate the implementation of forest management activities, and to invite investment in the forest area. In implementing forest management activities, the KPHP can delegate the management rights to concession holder from private and/or public (community).

¹According to the Ministry of Forestry Regulation P.47/Menhut-II/2013 regarding Guidance, Criteria and Standard for Forest Utilization in designated area on Protection Forest Management Unit and Production Forest Management Unit (KPHP)

1.4 Other Entities Involved in the Project

Organization name	Ministry of Environment and Forestry
Role in the project	Government body to co-develop the project and advise KPHP TBS on all aspects of project development, especially regarding governmental rules and permits.
Contact person	1) Is Mugiono, 2) Armunanto
Title	Director of Planning, Utilisation, and Establishment of Forest Management Area.
Address	Manggala Wanabhakti Building Jakarta-Indonesia
Telephone	+62 21 570 4501/ 570 4502/ 570 4503/ 570 4504
Email	is_mugiono@yahoo.com ; armunantods@yahoo.com

Organization name	Korea-Indonesia FMU/REDD+ Project Management Unit (PMU)
Role in the project	Co-Developer. Assist in aspects of project development related to preparation, kick-off, and preparation of verification and validation process
Contact person	1) Lee Seong Hwan 2) Andi Andriadi
Title	1) Korea Co-Director 2) Indonesia Co-Director
Address	Manggala Wanabhakti Building Jakarta-Indonesia Block I 5th Floor Wing B Room 511
Telephone	+62 21 5773 790
Email	leesh00317@gmail.com , andriadiandi@gmail.com

Organization name	Korea Indonesia Forest Center
Role in the project	Co-Developer. Assist in finalizing project documents for verification and validation process, as well as registry process.
Contact person	Moon Jongpil
Title	Project Manager
Address	Manggala Wanabhakti Building Jakarta-Indonesia Block IV 7th Floor Wing C Room 723
Telephone	+62 21 5785 2991 / +62 812 9739 9095
Email	bucge@nate.com

Organization name	Kampar REDD+ Consortium
Role in the project	A consortium consists of independent consultants and universities to assist PMU in project development kick-off; design and management of forest carbon inventory; lead and supervise collection of field data; lead the development of baseline; develop project document, registry process, and advise KPHP TBS on aspects of project development. The Kampar REDD+ Consortium is led by SG Consulting, Korindo Group and its members are CER Indonesia, University of Riau, Lancang Kuning University and Seoul National University.
Contact person	Roh Gyeong Min
Title	Deputy General Manager, SG Consulting, Korindo Group
Address	13 th Floor Wisma Korindo. Jl. MT Haryono Kav 62. Jakarta Indonesia.
Telephone	+62 21 797 5959 (Ext. 392)
Email	gmroh@korindo.co.id

1.5 Project Start Date

Issuance date of the Long-term Forest Management Plan (RPHJP) of the KPHP TBS is 16 December 2014. The document stated that the proposed project area is officially designated as location for implementation of REDD+ activity. Thus the date is selected as the project's start date.

1.6 Project Crediting Period

The project's crediting period is 20 years. The start date is 16 December 2014 and the end date of the crediting period is 15 December 2034

1.7 Project Scale and Estimated GHG Emission Reductions or Removals

The project is estimated to generate 24.8 Mt CO₂ of net emission removals during the project period. It is beyond the threshold for large project of annual emissions of 1,000,000 tCO₂, thus the project is categorized as a large project.

Project Scale	
Project	
Large project	√

Table 1. Estimated net emission removals throughout crediting period

Year	Estimated GHG emission removals (tCO ₂ e)
2015	316,244.59
2016	587,407.70
2017	805,952.62
2018	1,013,123.14
2019	1,167,906.65
2020	1,285,413.56
2021	1,364,257.11
2022	1,429,130.99
2023	1,419,873.39
2024	1,409,068.10
2025	1,406,732.34
2026	1,395,974.53
2027	1,388,319.68
2028	1,380,374.29
2029	1,429,130.99
2030	1,419,873.39
2031	1,409,068.10
2032	1,406,732.34
2033	1,395,974.53
2034	1,388,319.68
Total estimated ERs	24,818,877.73
Total number of crediting years	20
Average annual ERs	1,240,943.89

1.8 Description of the Project Activity

Main objective of the proposed project is to avoid conversion of the 14,722.9 ha of the proposed project area into timber plantation by assign it as a REDD area thus change its function from production to protection.

The area is part of “*wilayah tertentu*” (“specific area”) of the KPHP TBS. By regulation, the “specific area” is defined as area of KPHP in which no concessional rights have been awarded to any party. Within the KPHP TBS, there are about 40,000 ha “specific area”. The KPHP has the rights to manage this area for production of timber and/or non-timber forest products.

In line with the function of the KPHP as defined in Minister of Forestry Decree no. 509/Menhut-VII/2010, this area is planned to be managed for timber production,

which was also stated in the initial meeting for development of the KPHP TBS's management plan. However, considering the potential of the proposed area for forest preservation, the Government of Indonesia received an offer to have a joint cooperation in strengthening the capacity of forest management unit at Tasik Besar Serkap from the Government of Korea, in which one of the activity in to promote REDD+ and other carbon emission reduction mechanism. Following this, the area was then allocated for the cooperation to be conserved under REDD scheme. This plan was then formalized in the KPHP TBS's RPHJP (*Rencana Pengelolaan Hutan Jangka Panjang* - Long-term Forest Management Plan) and legalized by the Ministry of Forestry Decree No. 7565/Menhut-2014.

The proposed project will apply several measures designed to preserve the area from being deforested and/or experiencing degradation, such as:

- Installation of clear boundary that can be easily recognized by community
- Socialization with relevant stakeholders on the area's designation as REDD project site
- Facilitation on development of livelihood for surrounding communities, such as development of community forest.
- Collaboration with community under a program called *Masyarakat Peduli Api*² (community care fire) to manage the fire risk in the area².
- Empowering the community surrounding the project area to encourage the community participation and improve their concern to protect the forest ecosystem from degradation. The activities include facilitation to the development of small and medium businesses (UKM – *Usaha Kecil dan Menengah*) in the villages surrounding the KPHP-TBS, identification and development of business plan, as well as development of ecotourism activities, community rubber plantation and Sago community forest outside the proposed project boundary, forest-based honey bee, optimization of unproductive land in the villages for agrosilvopastural, and development of fish pond silvofishery.

1.9 Project Location

The proposed project site is located inside the State Production Forest Area of Tasik Besar Serkap's Forest Management Unit– Production. Administratively, the area is part of Sungai Apit Sub District, Siak District, Riau Province.

²The system works on voluntary basis. In participating villages, there will be trained volunteers to monitor for the existence of flames that potentially become fires. In the case that the flames are manageable, self-help extinguishing is then conducted; otherwise they will report it to officers in sub-district level and/or above.

According to the following map of project boundary from the KPHP, the project site is geographically lies between the coordinates 0°37'35,72" North - 0°45'57,38"North and 102°20'50,15" East - 102°38'19,29" East, as shown by the Figure 1 below.



Figure 1. The location of activity according to RPHJP KPH Tasik Besar Serkap 2014-2023

As shown by Land System Map from the Ministry of Agriculture (RePPProT, 1987). the proposed site is part of large peat ecosystem that stretches along the low land at east coast of Riau Province. Nevertheless, overlay analysis shows that the site located outside the peat moratorium zone as regulated by Decree of Ministry of Forestry no 6982 year 2014 (*Keputusan Menteri Kehutanan 6982/Menhut-VIII/IPSDH/2014*) or also known as PIPB 7³ (see Figure 2).

³ Presidential Decree No. 61/2011 has targeted reduction of GHG emissions by 26% of which 80% are from the LULUCF sector including peat. To support its implementation, GoI has issued Presidential Instruction No. 10/2011 dated May 20, 2011 on New Permit Delays and Improving of Governance of Primary Forest and Peatland. As a follow up measures, the Ministry of Forestry issued the Ministerial Decree No. SK.323/ Menhut-II/2011 dated June 17, 2011 concerning Indicative Map of New Permit Delays (*Peta Indikatif Pembukaan Ijin Baru/PIPB*) on Utilization of Forest Product, Utilization of Forest Area Area, and Conversion of Forest Area. In accordance to periodic revision of the map, the latest version of PIPB map (version 7) has been issued through the Minister of Forestry 6982 / Menhut-VII / IPSDH / 2014 dated 13 November 2014.

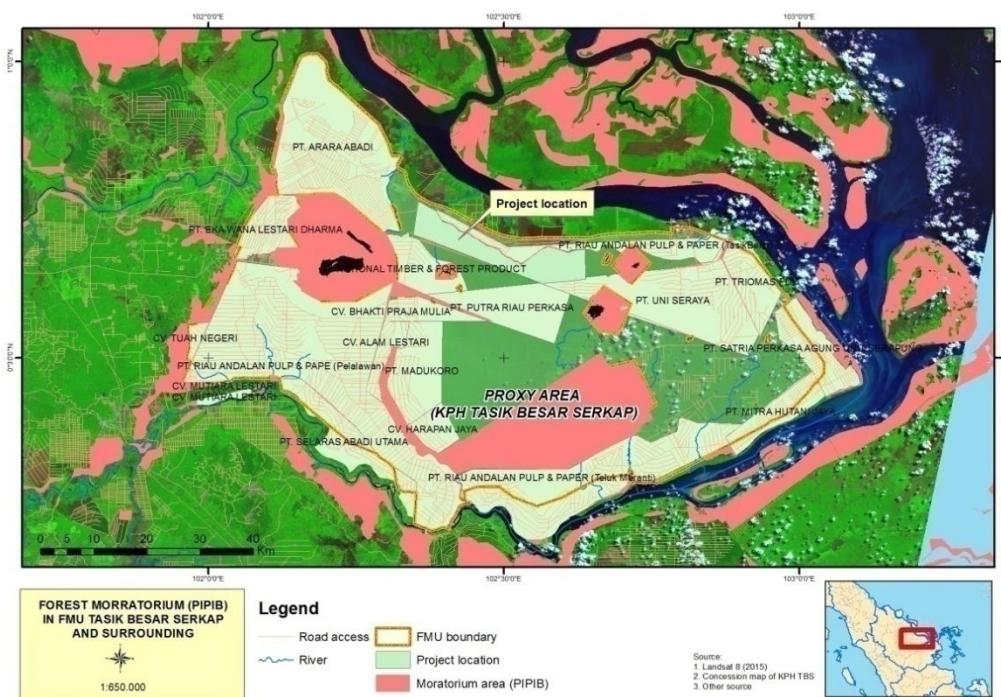


Figure 2. Overlay of project location and Peat Moratorium Map (PIPIB) Year 2014

1.10. Conditions Prior to Project Initiation

a. Physical and climate

Peat swamp forest in the project site is made up of young and old alluvium sediments consisting of sand sediment, lake, clay, peat and plant remains. Based on the national soil classification, the project site is dominated by Organosol / Peat soil which contains a lot of organic material, so that development of the land will be affected by the level of maturity, and the decomposition of organic material properties. Morphologically, the peat ecosystem is characterized by formation of soil horizons with dark brown to black colour, high water levels and very sour in acidity (pH 3-5). Based on the interpolation technique of field measurement data by using GIS, Peat depth varies, ranging from 4 to more than 12 m (see Figure 3)

Based on Digital Elevation Model (DEM) data from SRTM, the average altitude is 20 meters above sea level with slope ranging from 0 to 8% (flat area). Overall type of climate in the area by classification Schmidt - Fergusson is included as climate type A (wet climate) with average of annual rainfall in Siak (2011) is 2,443.80 mm. Temperatures ranging from 26 °C to 30 °C.

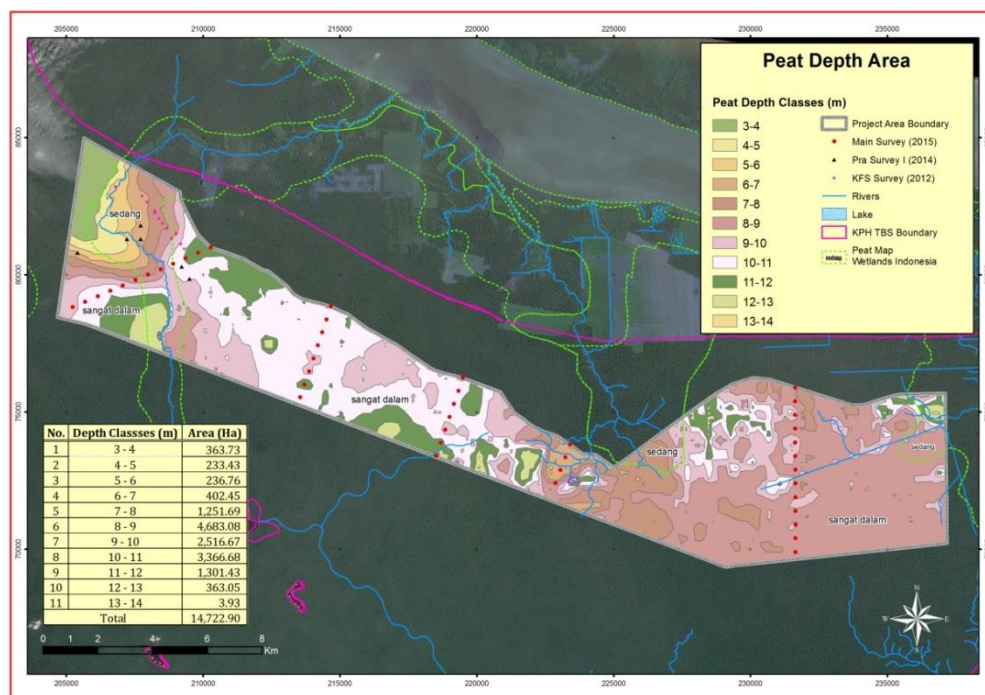


Figure 3. Peat depth assessment based on field survey

b. Landcover

Land cover of the project area was developed through Landsat 8/Operational Land Imager (OLI) classification. Overall accuracy of the classification was 90.79 % and Kappa statistic was 86.33%. Land cover in year 2014 was dominated by secondary peat swamp forest, with the coverage percentage reached 93% of the total acreage. The forest average is dominated by dipterocarpaceae trees. Other tree species found in the area are Kelat (*Syzygium aqueum* Burm.), Bintangur (*Callophyllum* spp), Gelam (*Melaleuca cajaputi* Powell), Jerejet, Kelincan (*Cinnamomum* spp), Mahakai, Pelintai (*Syzygium* spp), and Sepahit⁴.

Based on the Simpson diversity index, the proposed location falls into the category of complex and stable community. While based on the calculation of the Shannon-Wiener diversity index, level of diversity in the project area are classified as moderate category.

For the purpose of forest stratification with regard to above ground carbon stock measurement, the forest was differentiated further into four level of density, namely very low density, low density, medium density and dense forest. This will ensure the homogeneity of plot during measurement (Subedi et al., 2010). For that reason then the object based stratification method was performed based on field observation. Meanwhile for forest density differentiation, the stem count was selected as key

⁴Long Term Management Plan of Tasik Besar Serkap (RPHJP) 2014 - 2023

criteria (Kohl, Magnusen & Marchetti, 2006). In addition to that tree crown coverage and understory vegetation condition were also consider. The criteria and indicator of stratification are follows:

- *Very low density of forest* : secondary degraded forest which is dominated by sapling & poles
- *Low density secondary forest*: Forest cover in which within a 20 x 20 m plot, there are 20-40 trees stand, since the crown cover is relatively low, therefore the forest floor is mostly occupied by seedling, shrubs and pandanus (*Pandanus spp*).
- *Medium density forest*: Forest cover in which within a 20 x 20 m, there are 15-60 trees stand, whereas some trees have high size of diameter. Sapling and poles dominated in the forest floor. Very often tall pandanus (*Pandanus spp*) species can be found.
- *Dense Forest*: Forest with very high crown coverage, that make forest floor relatively clear compare to low and medium density. Within a 20 x 20 m plot dimension, there are 70 – 100 trees, in which some of them have very tall. Very often, very tall Mengkuang (*Pandanus odoratissimus*) exist.

These vegetated area cover 14,536.4 ha (99%) of the total area allocated for the project.

Bareland land cover types originated from secondary forests that have suffered severe degradation due to land clearing for agricultural purposes or as a result of forest fires. Barren land currently can be found in the western part of the area which already has good road access to facilitate encroachment by surrounding communities.

There is no settlement found inside the proposed project area.

The land cover map in the project area can be seen in Figure 4 below.

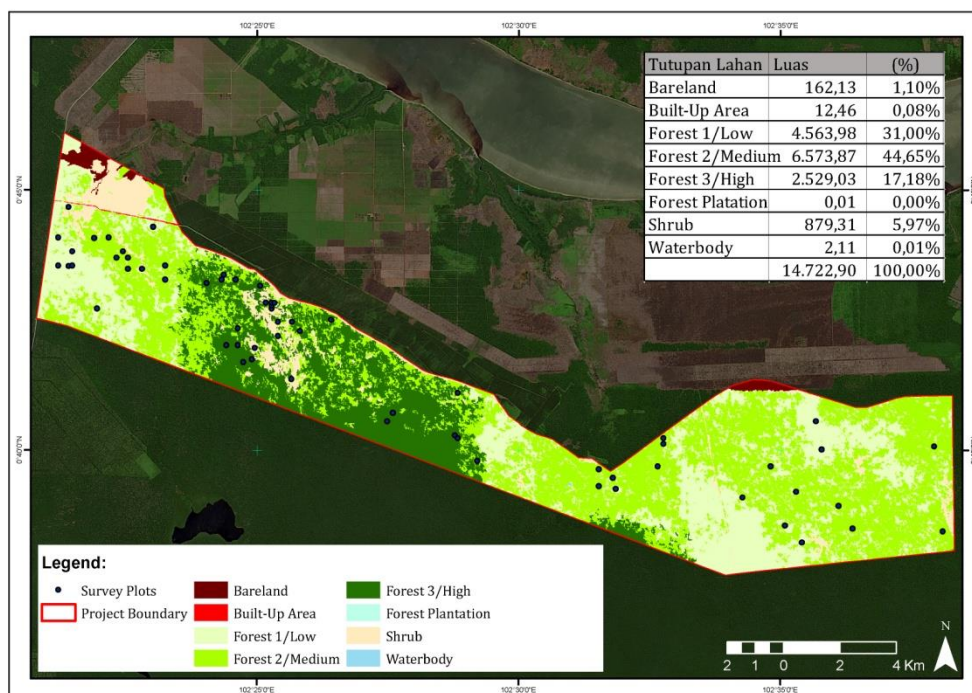


Figure 4. Landcover of project area year 2014 based on Landsat 8 Imagery

c. Biodiversity

Biodiversity assessment survey was conducted within the period of 16 March – 17 May 2015. The survey was divided into three stages, the first is pre survey conducted on 16 – 22 March 2015, the first main survey on 21 – 27 April 2015, and the last is second main survey on 8 – 17 May 2015. Furthermore, to support the existing data, a rapid assessment was conducted on 20 – 28 October 2015.

Flora

Based on the results of the survey, 67 species of timber plants and 29 species of non-timber plant were found. The level of biodiversity of flora in the area of REDD + Kampar including medium category with H' (Shannon Wiener Index) value of 2.99. Species richness exists in the area is considered as high category level with Margalef Index (R) of 8.97.

The most dominant species in the area are Kelat (*Syzygium aqueum*), 59.24%; Medang (*Cinnamomum spp*), 21.33%; Meranti (*Shorea spp*), 20.60%; Bintangur (*Calophyllum spp*), 20.10%; and Arang-arang (*Diospyros macrophylla* Bl), 15.49%.

Out of the identified plants species, Resak paya (*Vatica teysmanniana*) and Meranti bunga (*Shorea teysmanniana*) are native (endemic) of Sumatra, being found only in peat lands of South Sumatra, Riau and Bangka.

With regard to the IUCN Red List, one species of woody plants is categorized as Critically Endangered (CR), one species categorized as Vulnerable (VU) and one species categorized as Endangered (EN). In addition, there are also four non-timber plants species are categorized as Least Concern (LC).

Fauna

Fauna identified in the proposed project area is divided into three categories, namely species of herpetofauna, avifauna and mammals. Based on the survey, 28 species of herpetofauna, 75 species of avifauna, and 16 species of mammals were found.

Rapid assessment was conducted two transects (T1 and T3). Herpetofauna species dominance index shows differences in species composition in each transect. Out of the total of 18 species found, only five species are found in both transects. This is because the two transects are relatively distinct communities.

The biodiversity level of avifauna is considered as high category with H' (Shannon Wiener Index) value of 3.32 at T1 and 3.37 at T3. Both transects is dominated by a group of *Aerial feeders*. Overall, the group of birds most commonly found are insect eaters, followed by fruit-eating birds.

The level of biodiversity for mammals is at medium category with H' (Shannon Wiener Index) values of 2.29 in T1 and 1.92 in T3. The level of evenness (J) were high on both transects with a value of J at T1 0.76 and 0.61 at T3. There are differences in the level of encounter and the mammal's body size. The large mammals have higher percentage of encounter than the small-sized mammals.

Out of the of three identified species groups of fauna, Katak rawa kecil (*Hylarana rawa*), Perenjak jawa (*Prinia familiaris*), and Harimau sumatera (*Panthera tigris sumatrae*) are native (endemic) of Sumatra.

With regard to the IUCN Red List, for herpetofauna species, two are categorized as Vulnerable (VU) and 15 species are categorized as Least Concern (LC). In addition, there are nine species of herpetofauna categorized as Not Evaluated (NE), and 2 species (*Cyrtodactylus porosus* and *Sphenomorphus* sp.) are not included in the category of IUCN, CITIES, and the Indonesian government regulations.

For Avifauna, one species is categorized as Endangered (EN) and 14 species are categorized as Near Threatened (NT). For mammals, two species are categorized as Endangered (EN), two species are categorized as Near Threatened (NT), three species are categorized as Vulnerable (VU) and one species is categorized as Critically Endangered (CR). In addition, there are also six species of mammals categorized as Least Concern (LC) and one species categorized as Data Deficient (DD).

1.11 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project proponent is government entities eligible to develop REDD projects according to the Minister of Forestry Regulation P.20/Menhut-II/2012 regarding Implementation of Forest Carbon.

Since the proposed project area is a peat land with depth ranging between 5-12 m, there is a question on whether the area can be used for production use. With the issuance of the Government Regulation (PP) no. 71/2014 on Protection and Management of Peat Ecosystem, the Minister of Environment and Forestry should issue a decree to define the proposed area as protection function if it has peat land with depth of more than 3 meters and also has biodiversity function, protected species, and peat ecosystem.

However, application of the regulation needs further interpretation through the issuance of map indicating whether an area is assigned for protection or not, and based on the indicative map on moratorium for issuance of new permits prepared by the Ministry of Environment and Forestry, the proposed area is not included as moratorium area. Meaning that the function of the area remain as production area following the Ministry of Forestry decree No. 173/Kpts-II/1986 regarding the designation of the Tasik Besar Serkap area as production area. The indicative map is confirmed by the Minister of Forestry Decree no. 6982/Menhut-VII/IPSDH/2014 dated 13 November 2014. Thus the KPHP TBS have the rights to continue using the area for harvesting woods, including converting it to timber plantation forest.

Following the national policy related to the implementation of REDD as defined in the REDD National Strategy (2012)⁵, some of the production forest area located in peat land are included in the area to be conserved. In this regard, the proposed area is also included in the indicative map (PIPIB) dated 13th May 2015⁶ to confirm the intention of the KPHP TBS in allocating the area for REDD project.

1.12 Ownership and Other Programs

1.12.1 Right of Use

The land title of the area is a state owned forest area, according to the following legality:

- Minister of Forestry Decree Number 173/Kpts-II/1986 dated 6 June 1986 regarding Forest Land Use Agreement (TGHK), which was ammended by

⁵ Strategi Nasional REDD+, Satuan Tugas Persiapan Kelembagaan REDD+ Indonesia, Jakarta, Indonesia, Juni 2012. (<http://forestclimatecenter.org/files/2012-06-18%20Strategi%20Nasional%20REDD+.pdf>)

⁶ Legalized by Presidential Instruction no. 8/2015 regarding Moratorium of New Permit and Improvement of the Management of Proimary Natural Forest and Peat Land

Decree of Minister of Forestry Number 878/Menhut-II/2014 dated 29 September 2014 regarding State Forest Area of Riau Province.

- Riau Local Regulation (Perda) Number 10 year 1994 dated 19 August 1994 regarding Spatial Plan of Riau Province.

The state's owned forest area is managed by KPHP Tasik Besar Serkap (KPHP TBS).

1.12.2 Emissions Trading Programs and Other Binding Limits

According to the Indonesia's Action Plan for Greenhouse Gas Reduction, the establishment of Forest Management Unit (KPH) is designed to be a key strategy in achieving the national emission reduction target of 26% and 41%⁷. Although the proposed project is being developed under VCS scheme of which the credits generated can be sold to open buyer, the emission reduction generated from this project is also open to be used to fulfil the national target if it is not being sold to foreign buyer. Therefore, to avoid double counting, the proponent will send notification letter to the Ministry of Environment and Forestry who manages national registry on REDD Demonstration Activities as well as National Greenhouse Gas Inventory, and to Bappenas who tracks the progress of the national emission reduction target.

1.12.3 Other Forms of Environmental Credit

The project will only be registered under VCS standard.

1.12.4 Participation under Other GHG Programs

The project will only follow and registered to VCS standard.

1.12.5 Projects Rejected by Other GHG Programs

The project has not been registered to other forest carbon standard.

1.13 Additional Information Relevant to the Project

Eligibility Criteria

The proposed project area is single project, thus there is no eligibility criteria for inclusion of new instances of project activity

Leakage Management

⁷ The target is stipulated in the Presidential Regulation no. 61/2011 regarding National Action Plan for Greenhouse Gas Emission Reduction

Potential leakage from displacement of activities is considered zero therefore no activity addressing the possibility of shifting of activity is designed.

Commercially Sensitive Information

N.A

Further Information

N.A

2. APPLICATION OF METHODOLOGY

2.1 Title and Reference of Methodology

The proposed project is developed following the Approved VCS Methodology VM0007 ver.1.5 sectoral scope 14 titled REDD+ Methodology Framework.

The following modules and tools are used in the preparation of the VCS-PD, in accordance to guideline described in the VM0007 ver 1.5

- Module BL-PL for estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation and planned degradation, version 1.2
- Module CP-AB for estimation of carbon stocks in the above and below ground biomass in live tree and non-tree pools, version 1.1
- Module BL-PEAT for Estimation of baseline soil carbon stock changes and greenhouse gas emissions in peat land rewetting and conservation project activities; version 1.0
- Module E-FFC for Estimation of emissions from fossil fuel combustion; version 1.0
- A/R Methodological tool “Estimation of direct nitrous oxide emission from nitrogen fertilization”, version 01
- Module M-REDD/M-MON methods for monitoring of greenhouse gas emissions and removals in REDD project activities, version 2.1
- Module M-PEAT methods for monitoring of soil carbon stock changes and greenhouse gas emissions and removals in peatland rewetting and conservation project activities, version 1.0
- Module LK-ASP for calculation of leakage from shifting of activities, version 1.2

- Module X-UNC for Estimation of uncertainty for REDD+ project activities, version 2.1
- A/R Methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities”, version 01

2.2 Applicability of Methodology

The project activity conforms to the eligibility stated in the VM0007 as follows:

Table 2. Conformity of the project activity to eligibility criteria

No	Applicability point	Project situation
1	All land areas registered under the CDM or under any other carbon trading scheme (both voluntary and compliance-oriented) must be transparently reported and excluded from the project area	The proposed project area has no land registered under CDM or any other carbon trading scheme
2	Land in the project area has qualified as forest (following the definition used by VCS) at least 10 years before the project start date.	VCS forest definition is the same as CDM forest definition, therefore for Indonesian case the definition follow the Ministry of Forestry Regulation No 14/2004, which is also referred by the National Forest Reference Emission Level. Based on the regulation, forest definition is stated as follow: - Minimum land cover 30% - Wide area 0.25 Ha - Minimum height of tree 5 metres at mature stage.
3	Leakage avoidance activities shall not include: • Agricultural lands that are flooded to increase production (e.g. paddy rice); • Intensifying livestock production through use of “feed-lots” and/or manure lagoons	Leakage associated with planned deforestation baseline scenario is related to leakage due to shifting activity. Since no such leakage is estimated to occur, there is no leakage avoidance activities being planned
4	Conversion of forest lands to a deforested condition must be legally permitted	The proposed project area is a designated production forest area in which conversion into other forest form is permitted under the Ministry of Forestry decree No. 509/Menhut-VII/2010 on Establishment of the Tasik

		Besar Serkap Production Forest Management Unit (KPHP TBS)
5	Where, pre-project, unsustainable fuel wood collection is occurring within the project boundaries modules BL-DFW and LK-DFW shall be used to determine potential leakage.	No unsustainable fuel wood collection activities are identified prior to project implementation
6	REDD project activities on peat land shall not increase drainage	The REDD project activity is continuation of the current state of proposed project area as natural forest. No production activity will take place in the area thus the REDD project will not increase drainage

2.3 Project Boundary

2.3.1 Geographical boundary

Geographical boundary of the project comprising an area of 14,722.9 ha located in Siak and Pelalawan District. The proposed project area is surrounded by timber plantation forests (HTI). In the north side is timber plantation forest belongs to PT. RAPP (see Figure 5).

However, out of the 14,722.9 ha allocated for the REDD project, only 14,546.19 ha are being covered with vegetation while the remaining areas are in the form of water body and bare land, thus the vegetated area is the one use for estimation of carbon stocks and emissions.

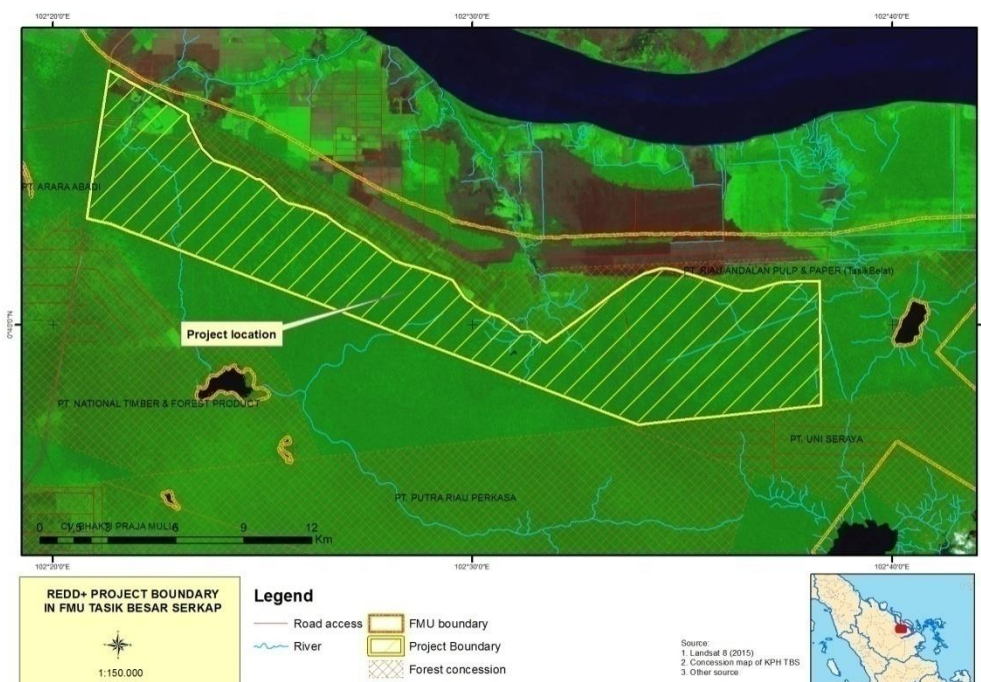


Figure 5. Proposed REDD+ Project Boundary

To demarcate field boundary of project site, KPHP Tasik Besar Serkap has installed permanent boundary markers (concrete pole/frame), which was conducted in October 2014. Installation of the boundary markers have been equipped with the minutes (*berita acara*) that describes the geographic coordinate of each boundary markers (see Appendix 1).

According to the Ministry of Forestry decree No. 509/Menhut-VII/2010 on, the proposed project area is eligible for conversion into other forest land uses such as industrial timber plantation, village forest, etc. At the time of the Project Description writing, there are already 17 Industrial Timber Plantation Forest, four Ecosystem Restoration (*Restorasi Ekosistem*), and two Village Forest (*Hutan Desa*) surrounding the area (see Table 3), making conversion of the area into industrial timber plantation the most plausible option.

Table 3. Existing Timber Plantation Concessions at KPHPTBS

No	Name of concession / parcels	Type	Permit and year	Area (ha)
1	PT. RAPP (consist of three separated location)	IUPHHK-HT ⁸	- SK. 248/MENHUT-VI/2009 - SK 180/Menhut-II/2013	141,795
2	PT. Mitra Hutani Jaya	IUPHHK-	SK 101/Menhut-II/2006	9,240

⁸ Permit for plantation

No	Name of concession / parcels	Type	Permit and year	Area (ha)
		HT		
3	PT. Satria Perkasa Agung	IUPHHK-HT	SK 60/Menhut-II/2013	11,927
4	PT. Triomas FDI	IUPHHK-HT	SK.522.21/IUPHHKHT/II/2003/012	9,625
5	PT. Uni Seraya	IUPHHK-HT	SK.214/Menhut-II/2007	33,360
6	PT. Putra Riau Perkasa	IUPHHK-HT	SK 104/Menhut-II/2006	15,640
7	PT. Madukoro	IUPHHK-HT	SK522.21/IUPHHKHT/II/2003/017	15,000
8	PT. Selaras Abadi Utama	IUPHHK-HT	SK 382/Menhut-II/2004	13,600
9	CV. Bhakti Praja Mulia	IUPHHK-HT	SK 522.21/IUPHHKHT/II/2003/11	5,800
10	CV. Alam Lestari	IUPHHK-HT	SK 522.21/IUPHHKHT/II/2003/15	3,300
11	CV. Harapan Jaya	IUPHHK-HT	SK 522.21/IUPHHKHT/II/2003/16	4,800
12	CV. Tuah Negeri	IUPHHK-HT	SK 215/Menhut-II/2007	1,480
13	CV. Mutiara Lestari	IUPHHK-HT	SK 522.21/IUPHHKHT/II/2003/007	4,000
14	PT. Arara Abadi	IUPHHK-HT	SK 703/Menhut-II/2013	45,431
15	PT. Ekawana Lestari Dharma	IUPHHK-HT	SK 733/Kpts-II/1997	9,300
16	PT. National Timber and Forest Product	IUPHHK-HT	SK 21/Menhut-II/2007	9,300
17	PT. Balai Khayang Mandiri	IUPHHK-HT	SK 20/Menhut-II/2007	22,250
18	PT. The Best One Uni Timber	IUPHHK-RE ⁹	Ongoing process for obtaining permit	44,112
19	PT. Gemilang Cipta Nusantara	IUPHHK-RE	SK 395/Menhut-II/2012	20,265
20	PT. Sinar Mutiara Nusantara	IUPHHK-RE	SK 162/Menhut-II/2014	37,100
21	PT. Global Alam Nusantara	IUPHHK-	SK 230/Menhut-II/2014	36,850

⁹ Permit for Ecosystem Restoration

No	Name of consession / parcels	Type	Permit and year	Area (ha)
		RE		
22	Desa Segamai Kec. T. Meranti	Village Forest	SK.154/Menhut-II/2013	2,000
23	Desa Serapung Kec. K. Kampar	Village Forest	SK 155/Menhut-II/2013	2,000

Source: RPHJP KPHP TBS, December 2014

2.3.2 GHG emissions sources associated with baseline and project scenario

The proposed project activity is avoid planned deforestation conducted in a peat land. As such, emission sources considered are those related to net stock changes and emissions occur due to deforestation activity in peat land.

Table 4. GHG emission sources in baseline scenario

Source	Gas	Included?	Justification/explanation
Stock changes in Above ground biomass	CO ₂	Yes	Tree and non-tree. Included as per VCS methodology VM0007 ver. 1.5 & VMD0006 BL-PL V1.2
Stock changes in Below ground biomass	CO ₂	Yes	Tree and non-tree. Included as per VCS methodology VM0007 ver. 1.5 & VMD0006 BL-PL V1.2
Stock changes in Soil (peat)	CO ₂	No	Changes in peat is covered in the estimation of emission from peat decomposition, thus this source is omitted
Stock changes in Litters	CO ₂	No	Optional, project proponent chose to not include it
Stock changes in Deadwood	CO ₂	No	From field survey, carbon stock of deadwood in the current land use (natural forest) is in average 8.6 ton C/ha, while under baseline scenario (timber plantation), deadwood is rarely found in the ground. Thus carbon pool post deforestation is less than pre-deforestation. Following the VM0007 ver. 1.5, the source is then omitted.
Stock changes in Wood products	CO ₂	No	Timber harvested in the area are most likely to be used for pulp and paper industry in which the wood

			products will be short-lived, thus emission from this source is omitted
Emission from Peat decomposition	CO ₂	Yes	The proposed project area is located in a peat land. Baseline scenario is estimate to cause peat decomposition thus emission from it is included.
Emission from Ditch drainage	CH ₄	Yes	Under the baseline scenario of managed tree plantation, ditches will be constructed to stabilize the hydrological system in the area thus emission from it is included.
Emission from Fuel consumption	CO ₂	Yes	Included as per VCS module VMD0006 BL-PL V1.2
Emission from Use of fertilizer	N ₂ O	Yes	Included as per VCS module VMD0006 BL-PL V1.2
Emission from Biomass burning	CH ₄	No	The practice of burning for land clearing in Industrial Timber Plantation is forbidden according to the Act no. 32/2009 on Environmental Protection and Management, thus it is assumed that biomass burning due to forest fire will not occur Moreover, Based on 15 year history of fire occurrence at Kampar Peninsula, there was no any single incidence within the project area (see Appendix 3)
Emission from Biomass burning	N ₂ O	No	

Table 5. GHG emission sources in project scenario

Source	Gas	Included?	Justification/explanation
Stock changes in Above ground biomass	CO ₂	Yes	Tree and non-tree. Included as per VCS methodology VM0007 ver. 1.5& VCS module VMD0006 BL-PL V1.2
Stock changes in Below ground biomass	CO ₂	Yes	Tree and non-tree. Included as per VCS methodology VM0007 ver. 1.5& VCS module VMD0006 BL-PL V1.2
Stock changes in Soil (peat)	CO ₂	No	The source is omitted in baseline scenario thus also omitted in project scenario
Stock changes in Litter	CO ₂	No	The source is omitted in baseline

			scenario thus also omitted in project scenario
Stock changes in Deadwood	CO ₂	No	The source is omitted in baseline scenario thus also omitted in project scenario
Stock changes in Wood products	CO ₂	No	The source is omitted in baseline scenario thus also omitted in project scenario
Emission from Peat decomposition	CO ₂	No	The project scenario is continuation of current forest and use as natural forest. No significant emission from peat decomposition occur on this state, thus this source is excluded
Emission from Ditch drainage	CH ₄	No	There will be no development of ditches under project scenario, thus this source is excluded

2.4 Baseline Scenario

Baseline scenario is conversion of the proposed project area to timber plantation forest. Most likely commodity developed under the timber plantation is *Acacia crassicaarpa*. Detail on the steps for selection of baseline scenario is described in the below Section 2.5 (Additionality).

Following the national policy on REDD, the objective of development of REDD in Indonesia is to save the country's natural forests and the flora and fauna within them. Thus conversion of natural forest into non-natural forest into for example, timber plantation, rubber plantation, and sago plantation is considered as deforestation. Since the proposed area is legally authorized for conversion to a man-made plantation, the activity proposed under this project is an avoided planned deforestation.

KPHP TBS is the agent of deforestation and it is estimated that deforestation under baseline scenario will occur at the rate of 1,037.94 ha/yr. Detail on the estimation of deforestation rate is presented in Section 3.1.1.

2.5 Additionality

According to the latest version of VCS Methodology: VM0007 REDD+ Methodology Framework (Version 1.5, 9th March 2015), the project's additionality should be assessed using the "Combined Tool to Identify the Baseline Scenario and Demonstrate Additionality in A/R CDM Project Activities" (version 01, 19 October 2007). As such, the following steps under the tool were applied:

STEP 1. Identification of alternative land use scenarios to the proposed project activity;

Sub-step 1.a. Identify credible alternative land use scenarios to the proposed project activity

Under the Ministerial Decree no. 509/Menhut-VII/2010, the proposed project area is a designated production area and KPHP TBS is given authority to manage it or to authorize it to third party (private and/or public entity).

Taking into account the abovementioned decree and land use practices in Kampar Peninsula area, there are seven options of land-use scenario that would have occurred on the proposed project area, i.e:

- A. Conversion to Forest Concession (*Hak Pengusahaan Hutan – HPH*) through licensing IUPHHK-HA¹⁰. Most likely the concession owner will implement selective cuttings.
- B. Conversion to Industrial Forest Plantation (*Hutan Tanaman Industri – HTI*) with *Acacia crassicarpa* as commodity through licensing IUPHHK-HT¹¹

As mentioned before, there are already 17 industrial timber plantations established in the KPHP TBS area, making it one of the alternatives to be developed in the proposed project area

- C. Conversion to Industrial Forest Plantation (*Hutan Tanaman Industri – HTI*) with Rubber as commodity through licensing IUPHHK-HT. During stakeholder process with surrounding community, some of them expressed their wish to have rubber plantation in the area
- D. Conversion to Community Plantation Forest (*Hutan Tanaman Rakyat – HTR*) through licensing IUPHK-HTR with sago as possible commodity

During discussion with surrounding community, some of them expressed wishes to have sago as commodity to be developed in the area. In order to accommodate the community's wish, development of sago can be seen as an

¹⁰ A Business License for the Utilisation of Forest Timber Products in Natural Forests hereafter abbreviated to IUPHHK-HA previously named Forest Concession (HPH) means a license to utilize production forests covering the activities of felling, transporting, planting, maintaining, securing, processing and marketing of forest timber products. (Based on Ministry of Forestry Regulation No P. 50/Menhut-II/2010 about the method granting and extending the work area of IUPHHK.)

¹¹ A Business Licenses for the Utilisation of Forest Timber Products in Plantation Forests at Production Forests hereafter abbreviated to IUPHHK-HTI previously named Plantation Forest Concession (HPHT) or Industrial Plantation Forest Concession (HPHTI) or a Business License for the Utilisation of Forest Timber Products at Plantation Forests (IUPHHK-HTI) means a business license to develop a plantation forest at a production forest which is developed by an industrial group to enhance the potential and quality of the production forest in the frame of meeting the demands for industrial basic material.

alternative, and the form of Community Plantation Forest is the most appropriate one since the land is a state land¹².

E. Conversion to Village Forest (*Hutan Desa* – HD), through licensing IUPHK-HD

There are already two Village Forests in the KPHP TBS area, making it feasible alternative

F. Designation as Ecosystem Restoration (*Restorasi Ekosistem* - RE) area, in which the concession will be given to management unit of a company applied and approved for the activity

Four Restoration Ecosystems have been established in the KPHP TBS area. Considering that the proposed project area is surrounded by plantation forest, concession holders wish to do conservation activity may apply to establish RE.

G. Avoidance of the area from conversion into other land use forms which is continuation of pre-project land use as natural forest

The KPHP TBS can conduct the abovementioned activities without applying for the specific license indicated in each of the activity. Rather, their mandate for management of the proposed project area is being stipulated in the Ministry of Forestry decision no. 509/Menhut-VII/2010 on the establishment of the KPHP TBS.

There is also palm oil plantation in some areas within the Kampar Bay area. However, legally this practice is prohibited since palm oil is classified as agricultural commodity which falls under the jurisdiction of the Ministry of Agriculture. Thus palm oil is not allowed to be developed in areas under the jurisdiction of the Ministry of Environment and Forestry.

Sub-step 1.b. Consistency of credible land use scenario with enforced mandatory applicable laws and regulations

The abovementioned seven options are all eligible under the Republic of Indonesia's existing laws and regulation. Thus, possible land use scenario resulted from application of Sub-step 1.b are similar to those resulted from application of Sub-Step 1.a.

STEP 2. Barrier Analysis

Sub-step 2.a. Identification of barriers that prevent the implementation of at least one alternative land use scenarios

Implementation of managed tree plantation may encounter the following barriers:

¹² The differentiation between Community Plantation Forest (*Hutan Tanaman Rakyat* – HTR) and Community Forest (*Hutan Rakyat* – HR) is that HTR is developed in state land while HR is developed in private land.

- a. Economic barrier, including the non-attractiveness of activity's economy, lack of access to initial capital, and lack of access to market
- b. Institutional barrier, such as the lack/weak institutional capacity on-site
- c. Ecological barrier, such as increase of forest fire due to improper water management

Possible barriers are identified and listed for each alternative land use scenario in the Table 9 below:

Table 6. Barriers identified in each alternative land use scenarios

Option	Barrier type
A. Concession area (IUPHHK-HA)	Considering the area which is only 14,722.9 ha (usually HPH area is more than 35,000 ha). Implementing HPH is not economically attractive
B. Acacia Forest Plantation (IUPHHK-HTI – <i>Acacia crassicaarpa</i>)	HTI will have negative impact on peat and also increase the risk of forest fire due to improper water management
C. Forest Plantation – Rubber (IUPHHK-HTI – <i>Hevea brasiliensis</i>)	Lack of access to market. HTI with rubber as commodity will have negative impact on peat and also increase the risk of forest fire due to improper water management. Also, since rubber requires intensive management, easy access to site is crucial.
D. Community Plantation Forest (HTR)	Lack of access to initial capital. Implementation of HTR also requires institutional capacity on-site which is lacking at the moment, thus requires institutional capacity development
E. Village Forest (HD)	Lack of access to initial capital
F. Restoration of Ecosystem (RE)	Economically not attractive
G. Avoidance of the area from conversion into other land use (continuation of existing land use)	Economically not attractive

Sub-step 2.b. Elimination of land use scenarios that are prevented by the identified barriers

All land use options have at least one barrier. However, there are other considerations overruling the barriers. The ecological barrier of HTI under scenario B is overlooked because of its economic attractiveness, while the barrier in scenario G is overlooked because of its benefit for preservation and GHG emission reduction.

Based on the application of sub-step 2.a. and the abovementioned rationale, the remaining alternative land use scenarios are:

- A. Conversion to Industrial Forest Plantation (*Hutan Tanaman Industri*– HTI) with *Acacia crassicaarpa* as commodity
- B. Avoidance of the area from conversion into other land use forms which is continuation of pre-project land use

Sub-step 2.c. Determination of baseline scenario

The proposed project's land use scenario without its inclusion in VCS REDD scheme is included in the list of land use scenarios in sub-step 2.b., therefore analysis move forward to Step 3.

STEP 3. Investment Analysis

The tool mentioned three options to determine the type of investment analysis to be used, i.e:

- Option 1 (simple cost Analysis) → applicable if the VCS AFOLU project generates no financial or economic benefits other than VCS related income.
- Option 2 (Comparison analysis)
- Option 3 (Benchmark Analysis)

Since the project generates no economical benefits aside than VCS related income, simple cost analysis is the one being applied

Sub-step 3.b. – Option 1. Apply Simple cost analysis

Three simple cost analysis were conducted on each of the following options

- A. Conversion to Industrial Forest Plantation (*Hutan Tanaman Industri*– HTI) with *Acacia crassicaarpa* as commodity;
- B. Avoidance of the area from conversion into other land use forms which is continuation of existing forest management; and
- C. Continued existing forest management (option B) under REDD scheme.

Option B and C do not involve new investment, therefore NPV was chosen for analysis instead of IRR.

Based on the Article 8 point 5 of the Ministry of Forestry Regulation No. 20/Menhut-II/2012 on Implementation of Carbon Forestry, 51% of GHG emission reductions or removals generated by forestry carbon project will be used to fulfil national target needs, the remaining 49% can be traded to foreign country. The project proponent

has options to sell the carbon credit to domestic buyer or international buyer. Following this, revenue considered in the calculation of investment analysis for option C is based on the selling of 49% of the estimated net emission reduction.

Calculation of the investment analysis is available for validation and summary of the analysis is presented below:

Option	NPV (IDR)
A	1,090,130,789,790
B	(53,831,338,216)
C	389,183,213,424

Baseline scenario is the scenario having the highest difference between incomes and costs over the crediting period. In this case is option A: conversion to Industrial Forest Plantation.

Sub-step 3.d. – Sensitivity analysis

Sensitivity analysis was conducted to see whether under different circumstances option A is still the most feasible option economically. There are two scenarios used in the sensitivity analysis, namely:

1. Decrease in direct costs (such as for maintenance, etc) of 10% per year while it is foreseen that the increase in fuel will cause an increase in operational costs at the rate of 10% per year
2. An increase in operational cost due to the increase in fuel price estimated at the rate of 20% per year. Meanwhile, the selling price of timber is decreasing at the rate of 20% per year.

Table 7. Result of sensitivity analysis

Option	NPV (IDR)	
	Scenario 1	Scenario 2
A. HTI Acacia	1,102,731,692,825	797,706,733,859
B. Avoidance of the area from conversion into other land use forms (as it is)	(59,214,472,038)	(64,597,605,859)
C. Avoidance of the area from		

conversion into other land use forms (REDD Scheme)	383,751,282,895	284,039,119,243
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Based on the result of the investment analysis, conversion to HTI-Acacia is the most economically attractive option thus serves as baseline.

STEP 4. Common Practice Analysis

The proposed project's land use activity is the continuation of pre-project land use which will avoid conversion of the proposed project area into other land uses. This land uses is not a common practice considering the proposed project area is within a designated forest production area and Industrial Forest Plantation (HTI) is the common land use in the area.

Based on the result of Step 1 to Step 4 in which:

- The proposed project activity is within compliance with law and regulation
- Implementation of proposed project activity has barrier which being overlooked due to its benefit for forest preservation and GHG emission reduction
- The proposed project activity is not the most economically attractive option
- The proposed project activity is not a common practice in the area

the proposed project activity is **additional**.

2.6 Methodology Deviations

The project apply the following deviations to the VM0007 ver. 1.5

According to module BL-PL ver. 1.2, in calculating the likelihood for deforestation "*likelihood (L-Di) will be equal to the proportion of similarly zoned proxy areas deforested within the previous five years within the appropriate stratum*". However, due to limited availability of cloud-free satellite imagery, the period used for the proxies is modified into within eight years since the proxy's issuance of permit. This period is chosen assuming deforestation activity was still taking place in the period.

The sources of peat emissions due to deforestation activity (under the baseline scenario) included in this report are CO₂ emission from peat decomposition and CH₄ emission from the drainage ditch. The emission factors for both and the estimate of area allocated for drainage ditches are based on IPCC (2014).

3. QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

3.1 Baseline Emissions

The baseline net GHG emissions for planned deforestation is determined as

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Where:

$\Delta C_{BSL,planned}$	: Net greenhouse gas emissions in the baseline from planned deforestation; t CO ₂ -e
$\Delta C_{BSL,i,t}$	: Net carbon stock in all pools in the baseline stratum i at time t ; t CO ₂ -e
$GHG_{BSL-E,i,t}$	: Greenhouse gas emissions as a result of deforestation activities within the project boundary in the baseline stratum i during project year t ; t CO ₂ -e/year
i	: 1, 2, 3, ... M strata
t	: 1, 2, 3, ... t^* years elapsed since the projected start of the REDD project activity

3.1.1 Calculating annual area of deforested land

Identification of the agent of planned deforestation in each baseline stratum i

The project site is located inside state-owned production forest area, with no significant restriction for converting the area into a timber plantation area. This shows that there is single planned deforestation agent which is timber plantation concession obtaining permit from the Ministry of Environment and Forestry.

Area of deforestation ($A_{planned,i}$)

According to official data from the Ministry of Environment and Forestry, there are 17 timber plantation concessions (IUPHHK-HTI) operating in the area of KPHP Tasik Besar Serkap. Historical data shows that prior to designation for REDD+ project, the project site had been allocated for timber plantation estates. Therefore, there is a threat of immediate deforestation in the project site. This threat can be demonstrated by the following evidences:

- Legal permissibility for conversion into managed tree plantation, including timber plantation

According to the Minister of Forestry regulation no. 509/Menhut-VII/2010, the KPHP TBS have authority to manage tree plantation activity in the area such as timber plantation, rubber plantation, and agroforestry.

- b. Suitability for conversion to alternative non-forest land use;

The proposed project area is part of the Kampar Peninsula peat land ecosystem. At the time of writing, there are 23 forest concessions permit holder in the KPHP TBS area, 17 of it are for industrial timber plantation activity (IUPHHK-HTI).

- c. Intent to deforest

Considering that the area's status is a production forest and one of KPHP's tasks is managing forest production, in the absence of the proposed project, the area is most likely to be converted into timber plantation in which deforestation will then occur.

Projection of Rate of deforestation ($D\%_{planned,i,t}$)

In accordance with the approved VCS methodology VM0007 ver 1.5, rate of deforestation of proposed project area can be projected through examining the rate of deforestation of certain proxy areas or average proportion of land that is cleared each year.

This "proxy area" approach is considered as the most suitable approach since there is no actual verifiable deforestation plan exists yet and the project area is not yet granted to any concessioners. The proxy areas are defined as areas under the management of KPH that met the following criteria:

- Land conversion practices shall be the same as those used by the baseline agent
- The post-deforestation land use shall be the same in the proxy areas as expected in the project area under business as usual
- The proxy areas shall have the same management and land use rights type as the proposed project area under business as usual
- Agents of deforestation in proxy areas must have deforested their land under the same criteria that the project lands must follow.

As previously mentioned, there are 23 forest concessions permit holder currently working inside the KPH area, from which 17 concessions are Timber Plantation (IUPHHK-HTI) permit, 4 concessions are Ecosystem Restoration (*Restorasi Ekosistem*), and 2 Village Forest (*Hutan Desa*).

From the 17 Timber Plantation Concession Permit, six concession parcels had been selected as proxy area for deforestation modelling, which were: 1) PT. Ekawana Lestari Dharma, 2) PT. Riau Andalan Pulp and Paper – Tasik Belat Block, 3) PT.

Mitra Hutan Jaya, 4) CV. Alam Lestari, 5) PT. Madukoro, and 6) CV. Harapan Jaya (see Figure 6 and Table 8).

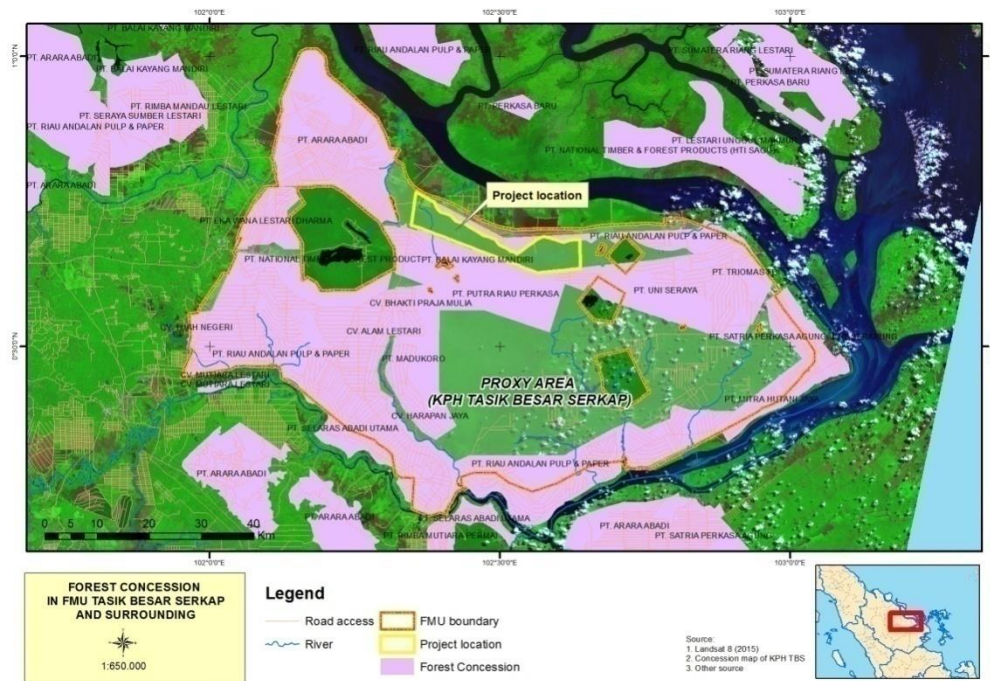


Figure 6. Forest consession in the area of the KPHP Tasik Besar Serkap

Those six proxy areas were expected to represent common land clearing practice in the proxy area, and hence also potentially occurred in the project area. Examination of proxy areas had been done mostly through remote sensing data (Landsat TM, Landsat ETM & Landsat OLI) classification analysis assisted by a set of verifiable spatial data from credible sources to support the analysis.

Table 8. Timber Plantation Concessions randomly selected as proxy areas

No	Name of concession holder	Type	Permit and year	Area (ha)
1	PT. RAPP (Tasik Belat Section)*	IUPHHK-HT	SK. 248/MENHUT-VI/2009	11,708.4
2	PT. Mitra Hutani Jaya	IUPHHK-HT	SK 101/Menhut-II/2006	9,240
3	PT. Madukoro	IUPHHK-HT	SK522.21/IUPHHKHT/I/2003/017	15,000
4	CV. Alam Lestari	IUPHHK-HT	SK 522.21/IUPHHKHT/II/2003/15	3,300
5	CV. Harapan Jaya	IUPHHK-HT	SK 522.21/IUPHHKHT/II/2003/16	4,800
6	PT. Ekawana Lestari Dharma	IUPHHK-HT	SK 733/Kpts-II/1997	9,300

**Section name is not official name from PT. RAPP, only used for this estimation purpose;*

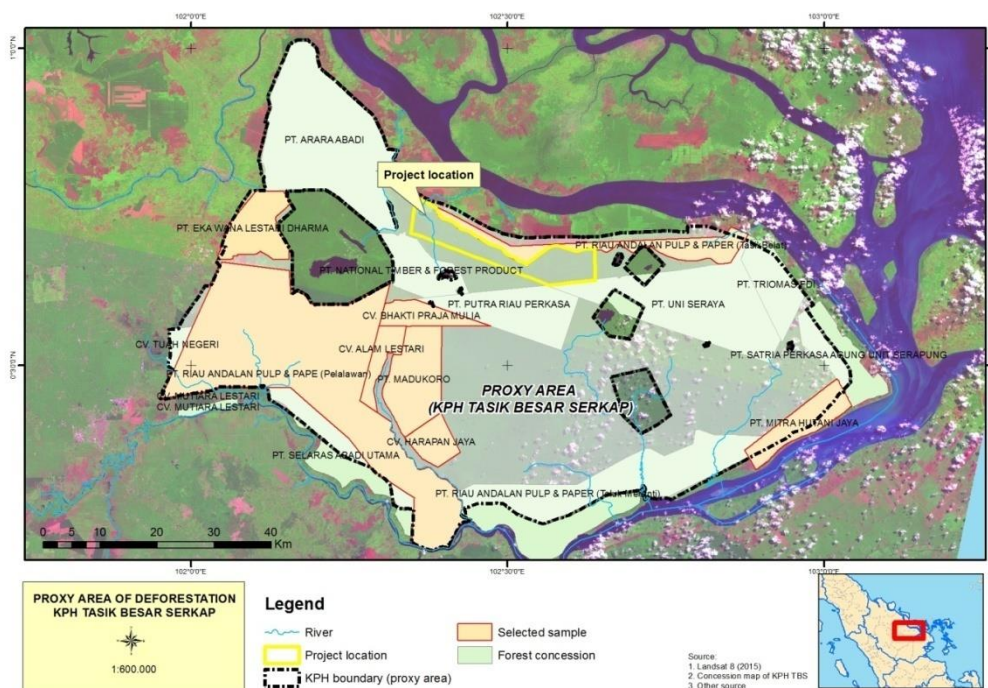


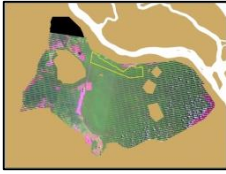
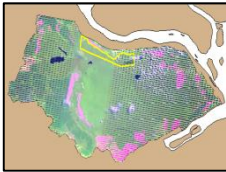
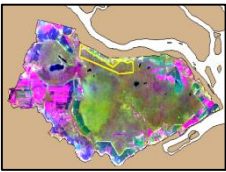
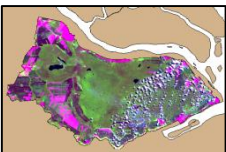
Figure 7. Proxy areas

For each selected proxy, a year to year land cover change analysis was performed with initial year (T_0) refers to each business license from the government. The earliest concession operation is PT. Ekawana Lestari Dharma (1997) and the most recent one is PT. RAPP (2009).

Differences in starting years of each sample of proxies' causes the analysis can not fully performed within period of 10 years due to limited availability of satellite data dating way back. In addition, proper satellite imagery was not always available for the expected year.

Table 9. Availability of Satellite imagery (Landsat) for landcover change

No.	Acquisition date	Preview
1	28 August 2001	
2	15 August 2002	

3	7 August 2005	
4	2 February 2009	
5	20 April 2011	
6	18 June 2013	
7	17 February 2014	

In defining area category in proxy area's satellite image, binary classification was applied, 1 for Natural Forest and 0 for non Natural Forest, including timber forest plantation (HTI), grasslands, shrubs, and barren land.

The projected rate of deforestation in the proposed project area is calculated with following equation:

$$D\%_{planned,i,t} = \left(\sum_{pn=1}^n \left(D\%_{pn} / Yrs_{pn} \right) \right) / n$$

Where:

$D\%_{planned,i,t}$: Projected annual proportion of land that will be deforested in stratum i during year t. If actual annual proportion is known and documented (e.g. 25% per year for 4 years), set to proportion (%)

$D\%_{pn}$: Percent of deforestation in land parcel pn etc of a proxy area as a result of planned deforestation as defined in this module; %

Yrs_{pn} : Number of years over which deforestation occurred in land parcel pn

in proxy area; years

n : Total number of land parcels examined

p_n : 1, 2, 3, ... n * land parcels examined in proxy area

i : 1, 2, 3, ... M strata

Result of calculation of the deforestation rate can be seen in the table 10 below.

Likelihood of deforestation ($L-D_i$)

The project area is under government control and the areas have been zoned for deforestation (production forest). The six parcels above had been examined to define the likelihood of deforestation occurring. Result of calculation of the likelihood of deforestation can be seen in the table 10 below.

Annual area of deforestation ($AA_{planned,i,t}$)

The projected annual area of deforestation in the baseline case was calculated as:

$$AA_{planned,i,t} = (A_{planned,i} * D\%_{planned,i,t}) * L - D_i$$

Where:

$AA_{planned,i,t}$: Annual area of baseline planned deforestation for stratum i at time t ; ha

$D\%_{planned,i,t}$: Projected annual proportion of land that will be deforested in stratum i during year t . If actual annual proportion is known and documented (e.g. 25% per year for 4 years), set to proportion; %

$A_{planned,i}$: Total area of planned deforestation over the baseline period for stratum i ; ha

$L-D_i$: Likelihood of deforestation for stratum i ; %

Result of calculation of the projected annual area of deforestation can be seen in the table 10 below:

Table 10. Land cover change and deforestation rate of proxy areas

No	Name of concession holders used for proxy	Year of permit granted	Concession area granted in the permit, in hectare (L)	Remaining area according to satellite analysis, in hectare						Total deforested area, D hectare	Number of years	Likelihood of Deforestation, (L-Di)	Deforestation rate %/year
				2001	2002	2005	2009	2011	2013				
1	PT. Ekawana Lestari Dharma	1997	9,300.00	7,828.11	6,044.32	3,198.67				4,629.44	4.00	59.14	14.8
2	PT. Madukoro	2003	15,000.00			14,782.20	8,573.56			6,208.64	4.00	42.00	10.5
3	CV Harapan Jaya	2003	4,800.00			4,995.40	2,852.65			2,142.75	4.00	42.89	10.7
4	CV Alam Lestari	2003	3,300.00			1,510.89	864.40			646.49	4.00	42.79	10.7
5	PT. Mitra Hutani Jaya	2006	9,240.00		8626.74	2,258.96				6,367.78	3.00	73.81	24.6
6	PT. RAPP - Tasik Belat block	2009	11,708.39				10,628.15	7390.82	5,543.23	5,084.92	4.00	47.84	12.0

Conclusion:

Total area of planned deforestation; A_{planned} , ha = 14,546.19

Projected annual proportion of land will be deforested; $D\%_{\text{planned}}$, (%) = 13.88

Average likelihood of deforestation; L-D (%) = 51.41

Annual area of baseline planned deforestation; AA_{planned} (ha/year) = 1,037.94

3.1.2 Carbon stock changes

Carbon pools included for calculation of net carbon stock changes are those from above ground and below ground tree and non-tree.

Following the module BL-PL ver. 1.2, net carbon stock changes is calculated as the pre-deforestation stocks minus the long-term post deforestation stocks. Each of the pre-deforestation and post-deforestation carbon stocks is calculated following below formula:

$$C_{P,i,t} = C_{AB_tree} + C_{BB_tree} + C_{AB_{non-tree}} + C_{BB_{non-tree}}$$

Where:

$C_{P,i,t}$	Carbon stock in all pools in the baseline case in stratum i at time t; t CO ₂ -e
$C_{AB_tree,i}$	Carbon stock in aboveground tree biomass in the baseline case in stratum i; t CO ₂ e
$C_{BB_tree,i}$	Carbon stock in belowground tree biomass in the baseline case in stratum i; t CO ₂ e
$C_{AB_non-tree,i}$	Carbon stock in aboveground non-tree vegetation in stratum i; t CO ₂ e
$C_{BB_non-tree,i}$	Carbon stock in belowground non-tree vegetation in stratum i; t CO ₂ e
i	1, 2, 3 ...M strata
t	1, 2, 3, ... t* years elapsed since the start of the REDD project activity

Factors used in calculating pre deforestation carbon stocks came from result of biomass survey conducted between March – May 2015 (see Table 11). While stratification was used in the survey and each stratum have its own stock factors, it is decided to use weighted average in the calculation because the absence of business plan for industrial timber plantation activity mean that there is no information on which exact strata is to be converted into plantation each year.

Assuming that each stratum will have the same possibility of being converted into timber plantation, calculation of weighted average carbon stock factor takes into account the proportion of each strata to total project area and total area planned for deforestation

As for the post-deforestation scenario, commodity used in the timber plantation forest is *Acacia crassiparva* and the value of its carbon stock factor is adapted from the research by Rochmayanto, *et al.* (2013).

Table 11. Biomass stock factors used in pre-deforestation stocks estimate

Above Ground tree	60.54 t C/ha
Above Ground non tree	1.40 t C/ha
Biomass Below Ground tree	6.05 t C/ha
Biomass Below Ground non tree	0.14 t C/ha

Results of the calculation are presented in below tables:

Table 12. Estimation of net carbon stock change in aboveground biomass

Year	Total C _{AB pre}	Total C _{AB post}	ΔC _{AB}	
	t C	t C	t C	t CO2
2015	497,992.54	433,955.56	64,036.98	234,802.27
2016	497,992.54	380,703.62	117,288.92	430,059.37
2017	497,992.54	340,039.77	157,952.78	579,160.18
2018	497,992.54	302,097.06	195,895.48	718,283.43
2019	497,992.54	276,687.13	221,305.41	811,453.17
2020	497,992.54	260,195.05	237,797.49	871,924.13
2021	497,992.54	252,959.30	245,033.24	898,455.21
2022	497,992.54	239,809.14	258,183.40	946,672.47
2023	497,992.54	242,023.88	255,968.66	938,551.77
2024	497,992.54	244,608.87	253,383.67	929,073.45
2025	497,992.54	245,167.67	252,824.87	927,024.53
2026	497,992.54	247,741.31	250,251.23	917,587.86
2027	497,992.54	249,572.61	248,419.93	910,873.08
2028	497,992.54	251,473.42	246,519.12	903,903.43
2029	497,992.54	239,809.14	258,183.40	946,672.47
2030	497,992.54	242,023.88	255,968.66	938,551.77
2031	497,992.54	244,608.87	253,383.67	929,073.45
2032	497,992.54	245,167.67	252,824.87	927,024.53
2033	497,992.54	247,741.31	250,251.23	917,587.86
2034	497,992.54	249,572.61	248,419.93	910,873.08
Total	9,959,850.83	5,435,957.88	4,523,892.95	16,587,607.48

Table 13. Estimation of net carbon stock change in belowground biomass

Year	Total C _{BB pre}	Total C _{BB post}	ΔC _{BB}	
	t C	t C	t C	t CO2

2015	11,528.87	10,639.64	889.22	3,260.49
2016	11,528.87	11,260.33	268.54	984.66
2017	11,528.87	13,643.34	(2,114.47)	(7,753.05)
2018	11,528.87	16,407.31	(4,878.44)	(17,887.63)
2019	11,528.87	20,925.87	(9,397.01)	(34,455.69)
2020	11,528.87	26,692.94	(15,164.07)	(55,601.58)
2021	11,528.87	33,755.88	(22,227.02)	(81,499.06)
2022	11,528.87	33,573.28	(22,044.41)	(80,829.51)
2023	11,528.87	33,883.34	(22,354.47)	(81,966.40)
2024	11,528.87	34,245.24	(22,716.37)	(83,293.37)
2025	11,528.87	34,323.47	(22,794.60)	(83,580.22)
2026	11,528.87	34,683.78	(23,154.91)	(84,901.35)
2027	11,528.87	34,940.17	(23,411.30)	(85,841.42)
2028	11,528.87	35,206.28	(23,677.41)	(86,817.17)
2029	11,528.87	33,573.28	(22,044.41)	(80,829.51)
2030	11,528.87	33,883.34	(22,354.47)	(81,966.40)
2031	11,528.87	34,245.24	(22,716.37)	(83,293.37)
2032	11,528.87	34,323.47	(22,794.60)	(83,580.22)
2033	11,528.87	34,683.78	(23,154.91)	(84,901.35)
2034	11,528.87	34,940.17	(23,411.30)	(85,841.42)
Total	230,577.38	579,830.17	(349,252.79)	(1,280,593.57)

Table 14. Net carbon stock changes

Year	Total stock pre-deforestation (t CO ₂)	Total stock post-deforestation (t CO ₂)	Total stock changes (t CO ₂)
2015	1,868,245.17	1,630,182.41	238,062.76
2016	1,868,245.17	1,437,201.14	431,044.03
2017	1,868,245.17	1,296,838.05	571,407.12
2018	1,868,245.17	1,167,849.37	700,395.80
2019	1,868,245.17	1,091,247.69	776,997.48
2020	1,868,245.17	1,051,922.62	816,322.55
2021	1,868,245.17	1,051,289.02	816,956.15
2022	1,868,245.17	1,002,402.21	865,842.96
2023	1,868,245.17	1,011,659.81	856,585.36
2024	1,868,245.17	1,022,465.09	845,780.08
2025	1,868,245.17	1,024,800.86	843,444.31
2026	1,868,245.17	1,035,558.67	832,686.51

2027	1,868,245.17	1,043,213.52	825,031.66
2028	1,868,245.17	1,051,158.91	817,086.26
2029	1,868,245.17	1,002,402.21	865,842.96
2030	1,868,245.17	1,011,659.81	856,585.36
2031	1,868,245.17	1,022,465.09	845,780.08
2032	1,868,245.17	1,024,800.86	843,444.31
2033	1,868,245.17	1,035,558.67	832,686.51
2034	1,868,245.17	1,043,213.52	825,031.66
Total	37,364,903.43	22,057,889.52	15,307,013.91

3.1.3 Emissions as a result of deforestation activity

Deforestation in the proposed project is caused by development of industrial timber plantation in peat land. Therefore the following sources of emissions are estimated:

- CO₂ emissions from fossil fuel combustion in the plantation's operation
- N₂O emissions from the use of fertilizer in the plantation
- CO₂ Emissions from decomposition of peat
- CH₄ emissions from drainage ditches

The methodology also mentions emissions due to biomass burning, typically generated during clearing for preparation of the land. However, practice in burning land for clearing is prohibited according to Act no. 32/2009 on Environmental Protection and Management thus this emission source is omitted.

Emissions from fossil fuel combustion

The baseline activity is estimated to according to the data from industrial plantation forest owned by PT. Korintiga Hutani that is also located in peat land. It used gasoline and diesel oil to fuel their operational vehicles (car, truck, motorcycle). Estimated fossil fuel consumption for each types of fuel is presented in Table 15.

Table 15. Fuel consumption per types of vehicle

Fuel type	Fuel consumption (liter/ha)
Gasoline	2.22
Diesel	47.17

Emission generated from the fossil fuel consumption is calculated using the following formulas:

$$E_{FC} = Fuel_a \times EF_a$$

$$Fuel_a = Liters_{Fuel,a} \times Density_{Fuel,a} \times NCV_{Fuel,a} \div 10^6$$

Where:

E_{FC} Net CO_{2-e} emissions of fuel consumption; t CO_{2-e}

$Fuel_a$ Amount of fuel of type a consumed; TJ

EF_a Emission Factor of fuel type a; tCO_{2-e}/TJ

a 1,2,3,... fuel types

$Liters_{Fuel,a}$ Quantity of fuel type a consumed; liter

$Density_{Fuel,a}$ Density of fuel type a; kg/liter

$NCV_{Fuel,a}$ Net Calorific Value of fuel type a; TJ/Gg

Default values such as NCV, and emission factors used for each types of fuel are following the 2006 IPCC Guideline and the Energy Statistic Manual as recommended in the module E-FFC ver. 1.0. Result of the calculation is presented below (see Table 16)

Table 16. Total emissions from fossil fuel consumption in 20 years

Vehicle type	Emissions (t CO2-e)
Gasoline	643.08
Diesel	16,045.16
Total	16,688.24

Emissions from the use of fertilizer

The amount of fertilizer used in *Acacia crassicarpa* planting is provided through estimation based on the common practice on timber plantation companies within Riau Province, Indonesia.

Table 17. Synthetic Fertilizer applied per type of fertilizer (application is only conducted one time per crop cycle)

Type	Synthetic Fertilizer applied (ton/ha)
Urea	0.34
ZA	0.39

NPK	0.5
-----	-----

Since no organic fertilizer is used, emission is only estimated from the use of synthetic fertilizer and calculated following the below formula:

$$N2O_{direct-N} = (F_{SN} + F_{ON}) \times EF_1 \times MW_{N2O} \times GWP_{N2O}$$

$$F_{SN} = \sum^I M_{SFi} \times NC_{SFi} \times (1 - Frac_{GASF})$$

Where:

$N2O_{direct-N}$	Direct N ₂ O emission as a result of nitrogen application within the project boundary; t-CO ₂ -e
F_{SN}	Mass of synthetic fertilizer nitrogen applied adjusted for volatilization as NH ₃ and NOX; t-N
M_{SFi}	Mass of synthetic fertilizer type <i>i</i> applied; tonne
EF_1	Emission Factor for emissions from N inputs; tonne-N ₂ O-N/(t-N input)
$Frac_{GASF}$	Fraction that volatilises as NH ₃ and NOX for synthetic fertilizers; dimensionless
MW_{N2O}	Ratio of molecular weights of N ₂ O and N (44/28); tonne-N ₂ O/(t-N)
GWP_{N2O}	Global Warming Potential for N ₂ O, kg-CO ₂ -e/kg-N ₂ O
NC_{SFi}	Nitrogen content of synthetic fertilizer type <i>i</i> applied; g-N/100 g fertilizer
<i>i</i>	Number of synthetic fertilizer types

Default values such as EF_1 , $Frac_{GASF}$, GWP_{N2O} are following the 2006 IPCC Guideline, while the value of NC_{SFi} is following Indonesian National Standard (Urea SNI 2801 2010; ZA SNI 02-1760-2005; NPK SNI 2803 2010). Result of the calculation is presented below (Table 18)

Table 18. Emissions from the use of fertilizer

Time scale	Emissions (t CO ₂ -e)
Per ha per application	1.17
Total per ha for 20 years (9 times application)	10.54
Per ha per year	0.53
Cumulative emission for 20 years	66,558.58

Emissions from decomposition of peat

Several assumptions were applied in the calculation:

1. All land covers (Forest 1, 2, 3, and shrub) in the 14,546.19 ha of project area are undrained under the initial condition (before the project starts) and thus the emission factor is 0 (IPCC 2006, IPCC 2014)
2. Emission factor of subsequent land use (Acacia plantation) if the conversion occur has a mean of 73.4 t CO₂ ha⁻¹ yr⁻¹ with 95% confidential interval, minimum value of 58.7 t CO₂ ha⁻¹ yr⁻¹ and maximum value of 88.1 t CO₂ ha⁻¹ yr⁻¹. Emission at this rate continues at a constant rate within the 20 years of project duration.
3. The rate of historical forest conversion is 1,037.94 ha/yr (see Section 3.1.1)
4. The calculation is made for 20 years period
5. Maximum conversion is 51.41% of the total project area, i.e. 7,478.2 ha. This conversion rate will be reached in 7.2 years since the deforestation activity started

Estimation of emission from peat decomposition is calculated following IPGG guideline in which activity data (annual area projected to be deforested; ha) is calculated with emission factor. For this calculation, emission factor of 73.4 t CO₂/ha/yr from the IPCC (2014)¹³ is used.

Emissions from drainage ditches

CH₄ emissions from drainage ditch follow was also calculated following IPCC in which activity data (annual area projected to be deforested; ha) is calculated with emission factor. For this calculation, emission factor of 2.259 tCH₄/ha/yr based on the IPCC (2014) is used.

An assumption of 2% deforested area is allocated for construction of ditch per hectare is used following the IPCC Supplement for Wetlands (IPCC, 2014).

¹³ IPCC, 2014. 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands. Hiraishi, T., Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J., Fukuda, M. and Troxler, T.G. (eds). IPCC, Switzerland

3.1.4 Summary of emissions/removals

Year	Carbon stock changes (t CO ₂)		Emissions due to deforestation activity (t CO ₂ e)				Total (t CO ₂ e)
	Aboveground biomass	Belowground biomass	Emission from consumption of fossil fuel	Emission from the use of fertilizer	Emission from peat decomposition	Emission from drainage ditches	
2015	234,802.27	3,260.49	136.93	547.07	76,184.80	1,313.04	316,244.59
2016	430,059.37	984.66	273.86	1,094.15	152,369.59	2,626.07	587,407.70
2017	579,160.18	(7,753.05)	410.79	1,641.22	228,554.39	3,939.11	805,952.62
2018	718,283.43	(17,887.63)	547.72	2,188.29	304,739.18	5,252.14	1,013,123.14
2019	811,453.17	(34,455.69)	684.65	2,735.37	380,923.98	6,565.18	1,167,906.65
2020	871,924.13	(55,601.58)	821.58	3,282.44	457,108.78	7,878.21	1,285,413.56
2021	898,455.21	(81,499.06)	986.62	3,829.51	533,293.57	9,191.25	1,364,257.11
2022	946,672.47	(80,829.51)	986.62	3,941.58	548,899.61	9,460.22	1,429,130.99
2023	938,551.77	(81,966.40)	986.62	3,941.58	548,899.61	9,460.22	1,419,873.39
2024	929,073.45	(83,293.37)	986.62	3,941.58	548,899.61	9,460.22	1,409,068.10
2025	927,024.53	(83,580.22)	986.62	3,941.58	548,899.61	9,460.22	1,406,732.34
2026	917,587.86	(84,901.35)	986.62	3,941.58	548,899.61	9,460.22	1,395,974.53
2027	910,873.08	(85,841.42)	986.62	3,941.58	548,899.61	9,460.22	1,388,319.68
2028	903,903.43	(86,817.17)	986.62	3,941.58	548,899.61	9,460.22	1,380,374.29
2029	946,672.47	(80,829.51)	986.62	3,941.58	548,899.61	9,460.22	1,429,130.99
2030	938,551.77	(81,966.40)	986.62	3,941.58	548,899.61	9,460.22	1,419,873.39
2031	929,073.45	(83,293.37)	986.62	3,941.58	548,899.61	9,460.22	1,409,068.10
2032	927,024.53	(83,580.22)	986.62	3,941.58	548,899.61	9,460.22	1,406,732.34
2033	917,587.86	(84,901.35)	986.62	3,941.58	548,899.61	9,460.22	1,395,974.53
2034	910,873.08	(85,841.42)	986.62	3,941.58	548,899.61	9,460.22	1,388,319.68
Total	16,587,607.48	(1,280,593.57)	16,688.24	66,558.58	9,268,869.18	159,747.82	24,818,877.73

3.1.5 Frequency of baseline renewal

The baseline estimation will be renewed once within the crediting period, i.e: in the year 2024.

3.2 Project Emissions

The proposed project is continuation of existing land use of natural forest so there will be no deforestation activity expected to occur. In terms of emissions from peat, the emission from peat decomposition is assumed zero since the area is still in its natural state, while areas surrounding the proposed project have been converted into timber plantation. Thus the change of condition outside of project area in the future might not be significant.

The potential factor that may affect the water condition within the proposed project area is development of road which is now still underway (see Figure 8). However, based on expert judgement (see Appendix 2), this may not have significant influence on water table in the project area. This is because construction of the road will involve the process of compaction of peat, so that the permeability of the peat will be very low. The construction of the road will have positive impact on water retention. Water drain from the area will be channelled to existing river networks.

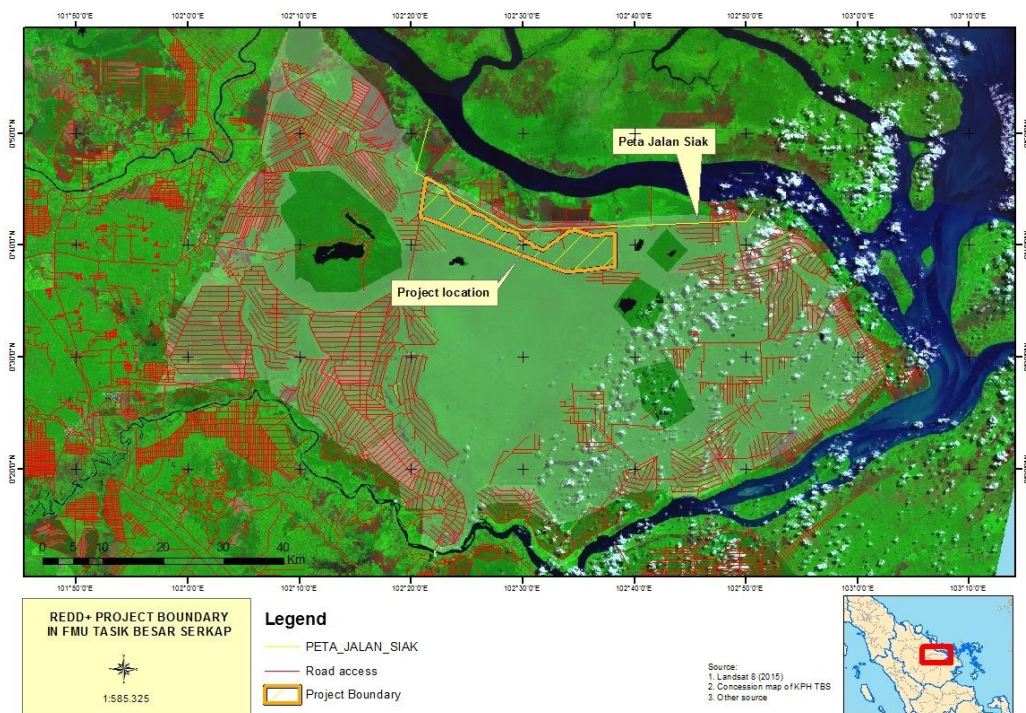


Figure 8. Plan for road development near proposed project area

3.3 Leakage

According to VCS Methodology VM0007, mandatory source of leakage to be considered under avoid planned deforestation project is leakage related to shifting of activity.

Deforestation agent in this project is the KPHP TBS, therefore leakage analysis is analyse on possible timber plantation activity the KPHP TBS might do to replace the one that will be conducted in the proposed project area in the absence of the REDD project.

In this regard, potential leakage from displacement of activities is considered to be zero since (a) all areas under the KPHP-TBS aside from the proposed project area is already being designated as production area and will remain the same, there will be no area under the KPHP TBS that function will be changed (e.g. from protection to production); and (b) the KPHP TBS have no authority to develop timber plantation outside of its assigned area according to the Minister of Forestry decree no. 509/Menhut-VII/2010 regarding the establishment of the Tasik Besar Serkap Production Forest Management Unit.

3.4 Net GHG Emission Reductions and Removals

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
1	316,244.59	0	0	316,244.59
2	587,407.70	0	0	587,407.70
3	805,952.62	0	0	805,952.62
4	1,013,123.14	0	0	1,013,123.14
5	1,167,906.65	0	0	1,167,906.65
6	1,285,413.56	0	0	1,285,413.56
7	1,364,257.11	0	0	1,364,257.11
8	1,429,130.99	0	0	1,429,130.99
9	1,419,873.39	0	0	1,419,873.39
10	1,409,068.10	0	0	1,409,068.10
11	1,406,732.34	0	0	1,406,732.34
12	1,395,974.53	0	0	1,395,974.53
13	1,388,319.68	0	0	1,388,319.68
14	1,380,374.29	0	0	1,380,374.29
15	1,429,130.99	0	0	1,429,130.99
16	1,419,873.39	0	0	1,419,873.39
17	1,409,068.10	0	0	1,409,068.10

18	1,406,732.34	0	0	1,406,732.34
19	1,395,974.53	0	0	1,395,974.53
20	1,388,319.68	0	0	1,388,319.68
Total	24,818,877.73	0	0	24,818,877.73

With non-permanence risk buffer deduction, the eligible VCU is estimated below

Year	Estimated carbon stock changes	Estimated total emission due to deforestation	Estimated project emissions or removals	Estimated leakage emissions	Risk buffer of 15%	Estimated net GHG emission reductions or removals
	(tCO ₂)	(tCO ₂ eq)	(tCO ₂ eq)	(tCO ₂ eq)	(tCO ₂)	(tCO ₂ eq)
1	238,062.76	78,181.83	0	0	35,709.41	280,535.18
2	431,044.03	156,363.67	0	0	64,656.61	522,751.10
3	571,407.12	234,545.50	0	0	85,711.07	720,241.56
4	700,395.80	312,727.34	0	0	105,059.37	908,063.77
5	776,997.48	390,909.17	0	0	116,549.62	1,051,357.03
6	816,322.55	469,091.01	0	0	122,448.38	1,162,965.18
7	816,956.15	547,300.96	0	0	122,543.42	1,241,713.68
8	865,842.96	563,288.03	0	0	129,876.44	1,299,254.54
9	856,585.36	563,288.03	0	0	128,487.80	1,291,385.58
10	845,780.08	563,288.03	0	0	126,867.01	1,282,201.09
11	843,444.31	563,288.03	0	0	126,516.65	1,280,215.69
12	832,686.51	563,288.03	0	0	124,902.98	1,271,071.56
13	825,031.66	563,288.03	0	0	123,754.75	1,264,564.93
14	817,086.26	563,288.03	0	0	122,562.94	1,257,811.35
15	865,842.96	563,288.03	0	0	129,876.44	1,299,254.54
16	856,585.36	563,288.03	0	0	128,487.80	1,291,385.58
17	845,780.08	563,288.03	0	0	126,867.01	1,282,201.09
18	843,444.31	563,288.03	0	0	126,516.65	1,280,215.69
19	832,686.51	563,288.03	0	0	124,902.98	1,271,071.56
20	825,031.66	563,288.03	0	0	123,754.75	1,264,564.93
Total	15,307,013.91	9,511,863.82	-	-	2,296,052.09	22,522,825.64

4. MONITORING

4.1 Data and Parameters Available at Validation

Data / Parameter	$A_{planned}$
Data unit	Ha
Description	Total area of planned deforestation over the entire project lifetime
Source of data	Calculated
Value applied	7,478.2
Justification of choice of data or description of measurement methods and procedures applied	Following section 1.2 of the module BL-PL ver. 1.2
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$L-D$
Data unit	%
Description	Likelihood of deforestation
Source of data	Calculated
Value applied	51.41
Justification of choice of data or description of measurement methods and procedures applied	Following section 1.4 of the module BL-PL ver. 1.2
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$D\%_{planned,i,t}$
Data unit	%
Description	Projected annual proportion of land that will be deforested in stratum i at year t
Source of data	Calculated
Value applied	13.88
Justification of choice of data or description of	Following equation 2 of the module BL-PL ver. 1.2

measurement methods and procedures applied	
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$AA_{planned}$
Data unit	Ha
Description	Projected annual area deforested
Source of data	Calculated
Value applied	1,037.94
Justification of choice of data or description of measurement methods and procedures applied	Following equation 3 of the module BL-PL ver. 1.2
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$C_{AB, tree\ post\ (acacia)}$
Data unit	t CO ₂ -e/ha
Description	Carbon stock in aboveground biomass of <i>Acacia crassicarpa</i> trees in the baseline
Source of data	Calculated. Carbon stock factor is based on a research by Rochmayanto, et al. (2013)
Value applied	Varies in years
Justification of choice of data or description of measurement methods and procedures applied	The carbon stock is estimated by calculating the amount of areas planted each years with stock factor
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$C_{BB, tree\ post\ (acacia)}$
Data unit	t CO ₂ -e/ha
Description	Carbon stock in belowground biomass of <i>Acacia crassicarpa</i> trees

	in the baseline
Source of data	Calculated
Value applied	Varies in years
Justification of choice of data or description of measurement methods and procedures applied	The carbon stock is estimated by calculating the aboveground tree stock with a conversion factor (below ground ratio) of 0.16
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$C_{AB, non-tree post (acacia)}$
Data unit	t CO ₂ -e/ha
Description	Carbon stock in aboveground biomass non-tree in <i>Acacia crasscarpa</i> area in the baseline
Source of data	Calculated. Conversion factor is based on a research by Putri Ramadhan (2011)
Value applied	Varies in years
Justification of choice of data or description of measurement methods and procedures applied	The carbon stock is estimated by calculating the aboveground tree stock with a conversion factor (non-tree ratio) of 0.0658
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$C_{BB, non-tree post (acacia)}$
Data unit	t CO ₂ -e/ha
Description	Carbon stock in belowground biomass non-tree in <i>Acacia crasscarpa</i> area in the baseline
Source of data	Calculated
Value applied	Varies in years
Justification of choice of data or description of measurement methods and procedures applied	The carbon stock is estimated by calculating the aboveground non-tree stock with a conversion factor (below ground ratio)
Purpose of Data	Calculation of baseline

Comments	N/A
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Data / Parameter	<i>R</i>
Data unit	t root d.m. t-1 shoot d.m.
Description	Root to shoot ratio applied as belowground biomass per unit area: aboveground biomass per unit area (not on a per stem basis)
Source of data	Hairiah K, Rahayu S. 2007. .Pengukuran “karbon tersimpan” di berbagai macam Penggunaan lahan. Bogor. World Agroforestry Centre – ICRAF, SEA Regional Office, University of Brawijaya, Indonesia.
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	<i>Carbon fraction</i>
Data unit	dimensionless
Description	Ratio applied as carbon stock: biomass stock
Source of data	SNI no.7224/2011
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	<i>AGB_{tree}</i>
Data unit	t C/ha
Description	Carbon stock factor for aboveground biomass tree in natural forest

Source of data	Estimated following field survey data
Value applied	60.54
Justification of choice of data or description of measurement methods and procedures applied	Stratification was used during field survey. The carbon stock factor is then a weighted average of stock factors in these stratum
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$AGB_{non-tree}$
Data unit	t C/ha
Description	Carbon stock factor for aboveground biomass non-tree in natural forest
Source of data	Estimated following field survey data
Value applied	6.05
Justification of choice of data or description of measurement methods and procedures applied	Stratification was used during field survey. The carbon stock factor is then a weighted average of stock factors in these stratum
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	BGB_{tree}
Data unit	t C/ha
Description	Carbon stock factor for belowground biomass tree in natural forest
Source of data	Estimated following field survey data
Value applied	1.4
Justification of choice of data or description of measurement methods and procedures applied	Stratification was used during field survey. The carbon stock factor is then a weighted average of stock factors in these stratum
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$BGB_{non-tree}$
Data unit	t C/ha
Description	Carbon stock factor for belowground biomass non-tree in natural forest
Source of data	Estimated following field survey data
Value applied	0.14
Justification of choice of data or description of measurement methods and procedures applied	Stratification was used during field survey. The carbon stock factor is then a weighted average of stock factors in these stratum
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	EFC_t
Data unit	t CO ₂ -e
Description	Emission from fuel combustion in year t
Source of data	Calculated
Value applied	Varies by years
Justification of choice of data or description of measurement methods and procedures applied	Following equation 1 of the module E-FFC ver. 1.0
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$Fuel_{a,t}$
Data unit	TJ
Description	Amount of fuel type a consumed in year t
Source of data	Calculated
Value applied	Varies according fuel type
Justification of choice of data or description of measurement methods and procedures applied	Following equation 2 of the module E-FFC ver. 1.0

Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$N_2O_{direct-N,i,t}$
Data unit	t CO _{2-e}
Description	Direct N ₂ O emission as a result of nitrogen application on the alternative land use within the project boundary in year t
Source of data	Calculated
Value applied	Varies by years
Justification of choice of data or description of measurement methods and procedures applied	Following equation A/R Methodological tool "Estimation of direct nitrous oxide emission from nitrogen fertilization"
Purpose of Data	Calculation of baseline
Comments	N/A

Data / Parameter	$E_{peatditch-CH_4,i,t}$
Data unit	t CO _{2e} ha ⁻¹ yr ⁻¹
Description	Emission of CH ₄ from ditch and open water stratum <i>i</i> at year <i>t</i>
Source of data	Project-specific values; appropriate literature sources
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	See module <i>M-PEAT</i>
Purpose of Data	Calculation of baseline emissions
Comments	N/A

4.2 Data and Parameters Monitored

Data / Parameter:	$C_{AB\ tree}$
Data unit:	t CO _{2e}
Description:	Carbon stocks of aboveground tree
Equations	Equation 29 of the module M-MON ver. 2.1

Source of data:	Calculated
Description of measurement methods and procedures to be applied:	A combination of ground measurement and satellite imagery analysis will be performed to estimate biomass stock. Carbon fraction will then convert the stock into carbon stock. The ground measurement will be carried out following the SNI no.7224/2011
Frequency of monitoring/recording:	Every three years
Value applied:	
Monitoring equipment	• Computer based GIS and remote sensing software • GPS for ground truthing • Camera
QA/QC procedures to be applied:	
Purpose of data:	Calculation of baseline
Calculation method:	• Change detection procedure using remote sensing tools • Overlay analysis of two landcover map using GIS equipment
Comments	

Data / Parameter:	$C_{AB \text{ non-tree}}$
Data unit:	t CO _{2e}
Description:	Carbon stocks of aboveground non-tree
Equations	Equation 29 of the module M-MON ver. 2.1
Source of data:	Estimated
Description of measurement methods and procedures to be applied:	A combination of ground measurement and satellite imagery analysis will be performed to estimate biomass stock. Carbon fraction will then convert the stock into carbon stock. The ground measurement will be carried out following the SNI no.7224/2011
Frequency of monitoring/recording:	Every three years
Value applied:	
Monitoring equipment	• Computer based GIS and remote sensing software • GPS for ground truthing • Camera
QA/QC procedures to be applied:	

Purpose of data:	Calculation of baseline
Calculation method:	• Change detection procedure using remote sensing tools • Overlay analysis of two landcover map using GIS equipment
Comments	

4.3 Monitoring Plan

Monitoring on carbon stock changes will be conducted for aboveground and below ground biomass. The proposed project will use both ground measurement and satellite imagery analysis to first estimate above ground biomass stock is the project area.

4.3.1 Forest cover change monitoring

The primary data used in forest cover change monitoring is high to medium data with 10 to 15 m spatial resolution. Landsat 8 imagery is the minimum imagery standard in combination with SPOT or higher resolution imagery.

A GPS device will be used to navigate the location of selected samples during ground verification. Digital camera will be used to photograph the land cover segments at visited sites for each class as reference data in the accuracy assessment.

The digital image processing will be carried out using computer based Remote sensing software for image processing, classification and analysis of the imagery.

Satellite image acquisition, pre-processing, secondary data collection, image classification, ground verification, accuracy assessment and output derivation are the main components in the monitoring process. Pre-processing should be performed to improve qualitative judgements concerning the data.

The classification system will be referred to the land cover map issued in the 2014 land cover mapping, which covers at least the following forest class: High density, medium density, low density, and very low density forest.

The data collected and posterior analysis shall cover:

- The entire reference region: at least at the beginning and at the end of the crediting period (for periodical revision of the baseline). Unit of analysis is ha yr⁻¹.
- The entire project area: at least at the beginning and at the end of each monitoring period (for periodical verifications). Unit of analysis is ha yr⁻¹.

- The entire leakage belt, where required: at least at the beginning and at the end of each monitoring period (for periodical verifications). Unit of analysis is ha yr⁻¹.

Two main change detection techniques can be used:

- Post-classification change detection: Two maps are generated for two different time points and then compared to detect changes in forest cover.
- Pre-classification change detection (direct classification of change): Two data sets acquired at different dates are analyzed together to detect the locations where a change has occurred

Automated classification methods should be used as the method of interpretation is more efficient and repeatable than a visual interpretation. Interpretation should also focus on change detection with interdependent assessment of two multi-temporal images together

An accuracy of the outcome of the mapping will be assessed by estimating the following three types of errors: (1) Overall classification accuracy (2) Error of omission of each category of change (error of excluding an area from a category to which it does truly belong, i.e. area underestimation) (3) Error of commission of each category of change (error of including an area in a category to which it does not truly belong, i.e. area overestimation) for each category of change, using statistical sampling

4.3.2 Analysis and Frequency

From forest cover change mapping, the following will also be covered in the analysis:

- Monitoring of areas undergoing degradation
- Monitoring of areas undergoing carbon stock enhancement

As stated in 2014 landcover map, the project contains forest areas that are assumed to be accumulating carbon and these areas are included in the baseline (Very low density, low density and medium density). Forest cover monitoring will be used for monitoring any change in the area of those mentioned strata.

A consistent time-series of land use-change will be used for periodical monitoring. Following standard of landcover monitoring in Indonesia, three years monitoring interval will be applied to achieve optimum result.

4.3.3 Ground measurement of biomass carbon stock

Ground measurement of biomass carbon stock in the project area will follow on procedure used during measurement for baseline pre-deforestation carbon stock. 63 permanent sample plots are

distributed within the project area. These sample plots were also those used for estimation of baseline pre-deforestation stock. Location of each sample plots is presented in Appendix 4.

Estimation of carbon stock will follow allometric equation by Murdiyarso (2010), thus ground measurement activity will only cover the measurement of diameter at each level of the stand (stakes, poles and trees) on various types of plants in the plots.

Calculation of carbon stock will follow the procedure used to estimate ex-ante pre-deforestation carbon stock. Below ground biomass stock will also be estimated using root-shoot ratio of 0.1¹⁴. Afterward, a ratio of 0.47¹⁵ will be used to convert the value of biomass stock (ton) into carbon stock (ton C).

5. ENVIRONMENTAL IMPACT

Proposed project activity is continuation of existing land use, i.e as natural forest, which does not require an environmental impact assessment according to the Minister of Environment No.5/2012 that Environmental Impact Assessments (EIAs) are required if the following conditions exist:

1. Concession Forest, all area scale
2. Timber Plantation, area is $\geq 5,000$ Ha

Therefore, EIAs is not required for the proposed activity as the proposed project activity is not developing a concession forest and timber plantation.

In addition, implementation of the project is expected to assist in conservation of biodiversity in the Kampar Bay area. As mentioned in the Section 1.10c regarding biodiversity, the proposed project area is home to 84 plant species, 23 herpetofauna, 75 avifauna, and 16 large mammals, some of it are endemic species such as Sumatran tiger (*Panthera tigris sumatrae*), avifauna (*perenjak jawa*), herpetofauna (*hylarana rawa*). There were also some endangered species such as agile gibbon (*owa unsko*), sunda pangolin (*trenggiling peusing*).

In terms of biophysical impact of the proposed project, no significant change is planned on the landscape as well as the composition of plants and peat since the project maintains current state of the land, i.e natural forest. Located in a peat land, the proposed project will avoid decomposition of peat that will otherwise occur if the area is converted into timber plantation.

¹⁴ Source: Hairiah K, Rahayu S. 2007. Pengukuran "karbon tersimpan" di berbagai macam Penggunaan lahan. Bogor. World Agroforestry Centre – ICRAF, SEA Regional Office, University of Brawijaya, Indonesia. (p. 32).

¹⁵ Source: Standar Nasional Indonesia – SNI (*Indonesian National Standard*) no. 7724/2011 regarding Measurement and Estimation of Carbon Stock – ground based forest carbon accounting. (p. 12).

With regard to community surrounding the area, social economy surveys were conducted first on February 2015 and second time on September 2015. Objectives of the survey was to collect information on the community's view on the REDD project, their social economy situation, their livelihood dependency on forest, and also to assist in formulating incentive for the community to reduce their dependency on forest products. According to the survey, more than 50% of the community's livelihood depends on forest products.

While implementation of the REDD project will limit their access to utilize some of the forest's products, it may also bring other opportunities, for example in the development of agroforestry and forest honey business.

6. STAKEHOLDER COMMENTS

On the proposed project, the KPHP TBS is the main proponent and stakeholders identified are:

- Five local communities living in the village surrounding the project area, namely the communities of *Desa Penyengat*, *Desa Rawa Mekar Jaya*, *Desa Sungai Rawa*, *Desa Mengkapan* and *Desa Teluk Lanus*.
- Concessions holders in the KPHP TBS area
- Riau Forestry Service
- Riau Agency of Planning and Development
- Ministry of Environment and Forestry

The project development process was begun since the signing of Record of Discussion (ROD) by two parties i.e. Korean and Indonesia Government on 25th January 2012. As a follow up, during the period of January 2012 – September 2015, the KPHP TBS has organized several meetings with local stakeholders in Riau in preparation of the REDD project implementation. For meetings with national stakeholders, some of it took place in Jakarta and Bogor.

Some of comments received during stakeholder consultation process are:

Private Plantation companies

- ☒ The organizational structure between the KPHP TBS and the concession holders should be improved in order to improve communication and information sharing between parties. The concession holders expect that in the future, the KPHP can provide leadership in facilitating inter-company interests.
- ☒ Concession holders in the area are supporting the implementation of REDD, as long as it is in line with government policy

Siak and Pelalawan Forestry Agency

- ☑ Since the KPHP is under direction of the Governor, the forestry agency of Siak district sees no direct relation with the district agency, instead only with provincial agency. As such, the district agency is not well informed of the KPHP's activities. Nevertheless, they support the implementation of REDD project

Local Communities

- ☑ Penyengat village communities has long story to develop sago. In 2014, they proposed a land area of 600 hectares for 300 families for planting sago. They wish that the KPHP can facilitate the development of sago plantation in their area.
- ☑ Rawa Mekar Jaya communities expressed their interest to plant rubber. They estimated that they need around 500 hectares of land that can be managed by 250 households. In the future, they hope the KPHP can facilitate the rubber development
- ☑ The local communities are supporting government's effort in preserving forest. However, they also want to note that the local economy' needs have to be taking into account.
- ☑ The community expect some clarity is given by the KPHP in terms of land used by the community. The rights given are utilization rights, and not ownership. Therefore, it will prevent any future conflict over land tenure.

Despite not having community living inside the proposed project area, the KPHP TBS will implements several community development program as part of preventive measures to avoid future encroachment in the proposed project area. In its RPHJP, they have planned to develop sago and rubber, thus the community's wish can then be accommodated in the form of development of community forest with sago and rubber as the main commodities. Rawa Mekar and Penyengat villages are chosen to be the location of these community forests.

APPENDIX 1. RESULT OF SATELLITE IMAGERY ANALYSIS ON FOREST FIRE OCCURRENCE IN THE PROJECT AREA

According to hot spot monitoring data from FIRMS (NASA), indication of hot spot cases within the area of the KPHP TBS in the last 15 years are as follows:

Year	Hot spot	Year	Hot spot	Year	Hot spot
2001	1	2006	53	2011	13
2002	28	2007	25	2012	41
2003	20	2008	20	2013	63
2004	48	2009	47	2014	451
2005	423	2010	4	2015	37

There were two hot spot peaks in years 2005 (423 cases) and 2014 (451 cases) with average of 85 hot spot cases every year. All of hot spot cases are located outside of project area, even though in 2014 some of them were occurred near the project location.

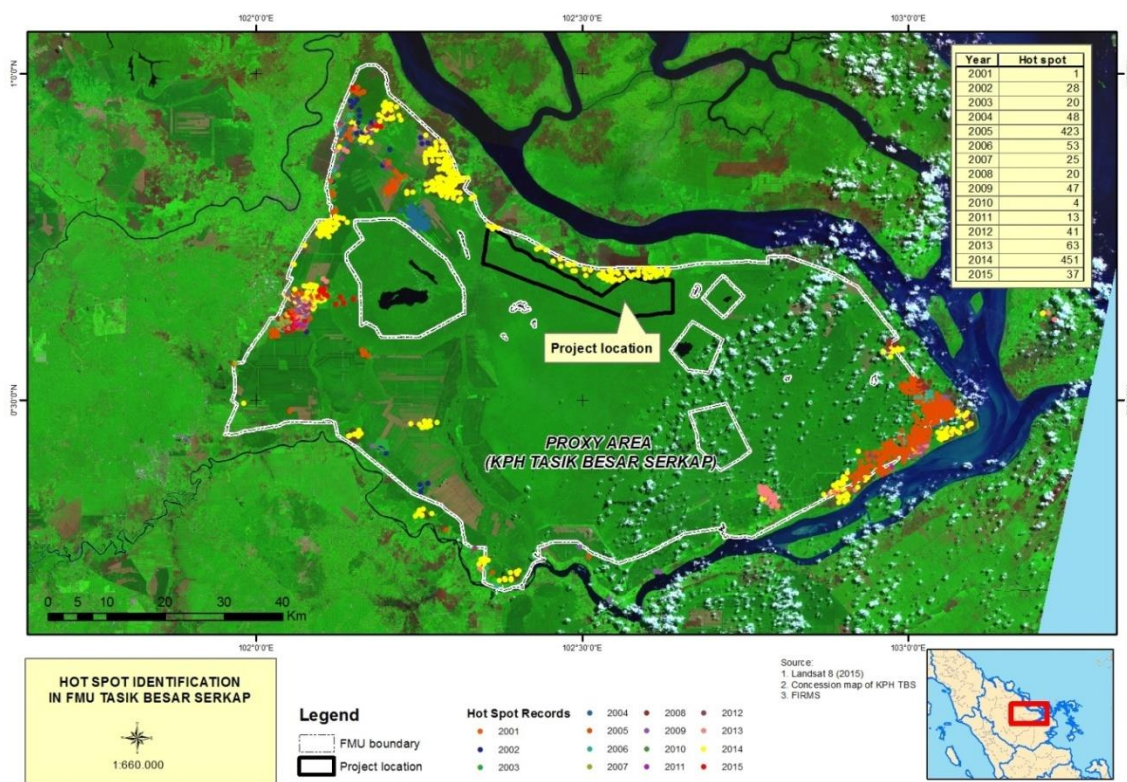


Figure 9. Map of Hot Spot Cases in Last 15 Years According to MODIS satellite (NASA)

APPENDIX 2. EXPERT JUDGEMENT ON THE CHANGE OF WATER TABLE WITHIN THE PROJECT AREA

Date	7 November 2015
Name of expert involved	Prof. Dr. Budi Indra Setiawan
Experts' background	Hydrologi Expert
The quantity being judged	The change of water table within project area under the baseline and project condition
The logical basis for judgement, including any data taken into consideration. This should include the rationale for the high end, low end, and central tendency of any uncertainty distribution	under the baseline condition (conversion of natural forest to industrial plantation with 7 years rotation) the emission from carbon stock change and GHG emission after the conversion only occur in the deforested area (converted area). In other word, the carbon stock in the remaining non converted area (natural forest) will not change significantly. Based on the expert judgement, the change may occur in GHG emission due to small change in water table (less than 20 cm on average in a year). Under project scenario, the change outside project area will be the construction of new road (see Figure 11). However, this may not have significant influence on water table in the project area. This is because construction of the road will involve the process of compaction of peat, so that the permeability of the peat will be very low. Instead, the construction of the road will have positive impact on water retention. Water drain from the area will be channelled to existing river networks.
The result: e.g., activity value, emission factor or for uncertainty the probability distribution, or the range and most likely value and the probability distribution subsequently inferred	Water table in the deforested area (HTI) ranging between 20-80 cm (on average 50 cm with 60% uncertainty), while in natural forest ranges between 0-40 cm (on average 20 cm).

APPENDIX 3. LOCATION OF PERMANENT SAMPLE PLOTS

